

Confidently Wrong: Exception Chain Collapse in Frontier LLM Rule Evaluation

A Neuro-Symbolic Architecture for Deterministic Eligibility Determination

Paul Simpson

John Kozak

Lisa Doake

July 2026

Abstract

We document a class of failure in frontier large language models — exception chain collapse — observed during eligibility evaluation under nested conditional rules of the form “A is required UNLESS B applies, UNLESS C overrides B”. The failure is real and reproducible at first observation, but its specific empirical surface is unstable: between March and April 2026 several v3.6/v3.7 paper cells closed silently under the same model alias, with no version bump (e.g. GPT-5.4 on construction-CAR moved from 96.6% to 100% on the same prompt and harness). For a regulated workflow that depends on benchmark-time accuracy claims, this is the central problem: the ground shifts under the production system without notice.

We present the Aethis Eligibility Module, a neuro-symbolic architecture that uses LLMs to author rules from authoritative sources, then executes those rules through an SMT-based deterministic evaluation layer. The Module achieves 100% consistency with the benchmark’s formal rule fixtures by construction — independent of model drift, reasoning-effort defaults, or prompt format. We harden this claim with three sources of evidence:

(i) *§6 controlled benchmark*. 225 scenarios across four regulatory domains document the original failure pattern (March 2026 snapshot). A v3.8 replication finds several v3.6/v3.7 cells have closed at current frontier performance; we report both numbers side-by-side as the model-drift evidence base.

(ii) *§6.4.1 v3.8 adversarial extension*. 20 newly-authored construction-CAR scenarios stratified across five complexity dimensions (independent-prose-then-engine methodology to address the code-derived ground-truth critique). Result: the Eligibility Module is 20/20 (100%); Claude Opus 4.7 (Anthropic’s strongest model at evaluation time, April 2026) is 18/20 (90%); GPT-5.4 at default reasoning effort is 19/20 (95%) but uses zero reasoning tokens on every scenario — short-circuit answers; Sonnet 4.6 is 19/20. Three of four frontier-LLM configurations fail on the same scenario (the DE3/LEG3 carveback-gap edge case), consistent with a structural rather than per-model failure mode.

(iii) *§6.10 external validation on LegalBench*. On nine peer-reviewed LegalBench tasks (949 held-out cases), the Eligibility Module is significantly more accurate than each of `claude-sonnet-4-6` (combined McNemar’s $p < 0.001$), `claude-opus-4-7` ($p = 0.003$), and `gpt-5.4` ($p < 0.001$); the structural advantage is largest on multi-prong rule-application tasks (Δ up to +41 percentage points) and persists at a smaller, directionally consistent margin on randomly-sampled tasks chosen without fit inspection (cross-task-significant there against `gpt-5.4`; the all-nine-task combination is the load-bearing test).

The architectural contribution is to relocate the locus of uncertainty: from the inference boundary, where it is silent and continuous, to the specification boundary, where it is deliberate and audited. The execution layer is guaranteed by construction; the specification layer is governed by SME-validated test suites and an explicit bundle-versioning regime (§5.5, §9.4). This relocates the problem rather than eliminating it.

All numbers in this paper are reproducible from committed scripts and result JSONs. The benchmark scenarios, the rule encodings, the harness, and the per-call replication artefacts are released publicly.

Contents

1. Introduction	3
2. Summary of Contributions	4

3. Related Work	5
3.1 Formal and Computational Approaches to Legal Reasoning	5
3.2 LLM Benchmarks for Legal and Logical Reasoning	6
3.3 Systematic Limits of LLM Reasoning	6
3.4 Prompt Engineering and Its Limits	6
3.5 Neuro-Symbolic Architectures and LLM + Formal Method Hybrids	7
3.6 SMT-Based Constraint Evaluation	7
4. The Problem: LLMs and Nested Exception-Chain Evaluation	7
4.1 What Makes a Decision “High-Stakes”	7
4.2 Two Failure Patterns	8
4.3 Why False Negatives Matter	9
5. The Neuro-Symbolic Architecture	9
5.1 Design Principle	9
5.2 Phase 1: Automated Rule Authoring	9
5.3 Phase 2: The Eligibility Module	10
5.4 Why the Eligibility Module Excludes These Errors by Construction	10
5.5 What the Eligibility Module Guarantees - and What It Does Not	11
6. Accuracy Benchmark: Eligibility Module vs Eight LLMs	12
6.1 Methodology	12
6.2 Scenario Design	14
6.3 Results	15
6.4 Construction Insurance: DE3/DE5 Defect Exclusion (58 Scenarios)	16
6.5 Error Analysis	21
6.6 The Cost-Accuracy Trade-off	22
6.7 Robustness Analysis	22
6.8 Threats to Validity	24
6.9 Pre-Registered Replication (v3.7)	25
6.10 External Validation on LegalBench	26
7. Challenges in LLM-Guided Rule Synthesis	32
7.1 Iterative LLM-Assisted Authoring	32
7.2 Honest Assessment	32
7.3 Test-Driven Validation	32
8. Generalisation and Applicability	32
8.1 Domain-Agnostic Architecture	32
8.2 Domain Injection Points	33
8.3 Adjacent Verticals	33
9. Compliance and Regulatory Considerations	33
9.1 Auditability	33
9.2 Explainability and Human Oversight	34
9.3 Quality Gates	34
9.4 Model Drift and Bundle Certification	34
10. Limitations and Future Work	35
10.1 Current Limitations	35
10.2 Future Work	35
Acknowledgments	36

References	36
Technical Appendix A: Benchmark Methodology	38
A.1 Life in the UK Scenario Matrix	38
A.2 English Language Scenario Design	38
A.3 Spacecraft Scenario Design	38
A.4 Construction Insurance Scenario Design	38
A.5 LLM Prompt Structure	39
A.6 Reproduction	39
Technical Appendix B: Architecture Overview	39
Author Contributions	40
Competing Interests	40
Data and Code Availability	40

1. Introduction

In 1986, Sergot, Sadri, Kowalski, and colleagues published a landmark paper in *Communications of the ACM*: they had formalised the British Nationality Act 1981 as a Prolog logic program (1986). The paper demonstrated that statutory legislation - with its characteristic structure of routes, exemptions, and override provisions - could be faithfully encoded as Horn clauses with negation as failure, and evaluated with provable correctness. It was a foundational result in computational law: legislation is logic, and logic can be executed.

Forty years on, large language models can read the same Act, explain it in plain English, and answer broad legal questions with fluency that would have seemed extraordinary even five years ago. It is tempting to conclude that the formal encoding project is no longer necessary. Our benchmark tests this assumption directly on the same legislation Sergot et al. encoded, extended to three further domains, and finds it does not hold - at least not for the specific class of task where exception chains are nested three levels deep.

The failure is specific. On straightforward multi-route eligibility logic, frontier models perform well, achieving 100% accuracy on 43 English-language scenarios (95% Wilson CI [91.8%, 100%]). But when legislation introduces nested exception chains, accuracy degrades sharply and in a way that is insensitive to temperature, sample count, and the prompting strategies tested here (a single-prompt baseline and one enhanced variant; see §6.7). Claude Opus 4.6 scores 61/68 (89.7%, CI [80.2%, 94.9%]) on the spacecraft section. With 10 runs on each of the 7 failing scenarios — 70 trials in total — it produces zero correct answers. The Clopper–Pearson 95% one-sided upper bound on per-trial success probability is 4.19%: the failures are systematic, not stochastic. Attempts to close the gap via enhanced prompting trade false negatives for false positives: the enhanced prompt reduces net accuracy to 64.7% (CI [52.8%, 75.0%]) while introducing 20 false positives for the first time.

More precisely, the failure regime characterised here is *multi-prong compositional rule evaluation under nested exceptions*: tasks where multiple coordinate clauses must each be evaluated against a typed input and then combined across two or more levels of UNLESS-style exception structure. The “exception chain collapse” terminology is retained throughout the paper as the dominant failure pattern within that regime; the more precise scope phrase is what we mean wherever the dominant pattern is invoked.

We make three contributions:

1. **A failure pattern taxonomy.** We identify and characterise two systematic failure patterns in nested exception-chain evaluation: *exemption anchoring* (failure to evaluate alternative routes independently

when the primary route fails) and *exception chain collapse* (failure to correctly nest multi-level UNLESS logic). We demonstrate both are systematic across six models (three frontier, three production-tier) and two providers on this benchmarked class of task.

2. **A multi-domain benchmark.** We present 225 scenarios across four sections spanning two UK immigration requirements (life-in-the-UK knowledge and English language proficiency), a synthetic spacecraft certification statute, and a synthetic construction insurance wording modelled on London market DE3/DE5 clause structure - designed to isolate exception chain evaluation as the test variable. Benchmark scenarios are released as a public dataset. The specific accuracy figures reported here will evolve as models improve; the task structure and failure pattern constitute the durable contribution.
3. **The Aethis Eligibility Module.** We describe a neuro-symbolic architecture that uses LLMs for rule authoring and an SMT-based constraint evaluation engine for rule execution, achieving complete consistency with the benchmark’s formal rule fixtures across all 225 scenarios, with <1ms evaluation latency and near-zero marginal cost after compilation.

The paper is structured as follows. Section 2 summarises key findings. Section 3 reviews related work. Section 4 characterises the failure pattern. Section 5 describes the architecture. Section 6 presents the benchmark. Section 7 discusses challenges in LLM-guided rule synthesis. Sections 8, 9, and 10 address generalisation, compliance, and limitations.

2. Summary of Contributions

Artificial intelligence is entering high-stakes decision-making: immigration eligibility, safety certification, insurance underwriting, benefits entitlement, financial compliance. These are domains where errors have material consequences, where false negatives deny people their rights, and where explainability is mandatory.

We present the **Aethis Eligibility Module** (hereafter: the Eligibility Module), a neuro-symbolic engine that separates what LLMs do well from what requires formal guarantees. LLMs read authoritative sources and generate rules as structured code. The Eligibility Module compiles and evaluates those rules using an SMT-based deterministic evaluation layer, providing constraint evaluation with mathematically defined semantics and full auditability.

The architectural contribution is to relocate the locus of uncertainty: from the inference boundary, where it is silent and continuous, to the specification boundary, where it is deliberate and audited. This does not eliminate the underlying uncertainty; it makes it tractable — addressable through SME-validated test suites, explicit bundle versioning, and the L1/L2/L3 separation set out in Section 5.5.

Our benchmark of 225 scenarios across four sections tests six LLMs (three frontier, three production-tier) in the March 2026 pass, with Claude Opus 4.7 and GPT-4.1-mini added in the April 2026 v3.8 replication, against the Eligibility Module on the specific task of nested exception-chain evaluation. The Eligibility Module achieves complete consistency with the benchmark’s formal rule fixtures across all domains - a consequence of deterministic execution over the authored specification, not empirical tuning. On adversarial exception-chain scenarios, frontier models produce systematic false negatives: Claude Opus 4.6 returns the wrong answer on 10% of spacecraft scenarios, and in the March 2026 baseline evaluations all observed failures are false negatives - eligible applicants incorrectly rejected, valid claims incorrectly denied (later evaluation arms do surface false positives; see §6.5 Finding 2). Attempting to improve LLM accuracy via enhanced prompting trades false negatives for false positives: the enhanced prompt reduces false negatives from 7 to 4 but introduces 20 false positives, reducing net accuracy from 90% to 65%.

On the benchmarked class of nested exception-chain tasks, LLM errors are systematic enough that deterministic formal execution removes one entire class of error from the trust surface for high-stakes decisions where consistency with formal fixtures is required. This is not a claim that LLMs are unreliable in general; it is a

specific finding about a specific class of rule evaluation.¹

3. Related Work

This work sits at the intersection of four research traditions: formal and computational approaches to legal reasoning; empirical evaluation of LLM logical reasoning; the limits of prompt engineering as a reliability strategy; and neuro-symbolic architectures that combine LLM fluency with formal execution guarantees.

3.1 Formal and Computational Approaches to Legal Reasoning

The challenge of encoding legislation as executable logic has a forty-year history. Sergot et al. (1986) formalised the British Nationality Act 1981 as a Prolog logic program, demonstrating that statutory rules could be faithfully represented as Horn clauses with negation as failure. Published in *Communications of the ACM* in 1986, this work identified the same legislation evaluated in our benchmark and showed that OR-branching eligibility logic can be captured in a formal system with provable properties. The Eligibility Module revisits the same statutory text with a different technical foundation - formal constraints compiled from LLM-authored rules rather than hand-coded Prolog - and extends evaluation to adversarial exception chain scenarios not part of the original formalism.

McCarty’s TAXMAN system (1977) demonstrated as early as 1977 that AI systems could reason over tax code with explicit logical representations. Bench-Capon and colleagues developed value-based argumentation frameworks for legal reasoning over multiple decades (2010), establishing that legislation’s exception structure requires more than propositional logic to represent faithfully.

The formal treatment for exception chains specifically is defeasible logic, introduced by Reiter (1980) and developed by Nute (1994) and Governatori et al. (2010). Defeasible logic provides formal semantics for “A holds UNLESS B applies, UNLESS C overrides B” - precisely the pattern our benchmark identifies as a failure pattern for LLMs. The Eligibility Module does not use defeasible logic directly, instead compiling exception chains to formal material implication constraints evaluated by an SMT solver, but operates in the same tradition: the failure pattern we document is exactly the problem defeasible logic was designed to solve, now re-emerging in systems that replaced formal encoding with statistical inference over legislative text.

The broader “Rules as Code” movement — encoding legislation as machine-executable representations at the point of drafting — has gained institutional momentum since 2020, with government-backed programmes in New Zealand, France, and Germany deploying visual decision-tree modelling tools for legislative analysis (Mohun and Roberts 2020; Mowbray et al. 2023). These systems encode statutory logic as structured flowcharts or decision trees authored via no-code configuration, enabling non-technical legislative drafters to model requirements directly. Mowbray et al. (2023) identify scalability as the central challenge: decision trees grow combinatorially as exception depth increases, and maintaining consistency across large rule sets requires manual audit. The approach shares this paper’s core premise — that statutory rules should be formally encoded rather than statistically inferred — but differs in representational power. Decision trees can model branching logic; constraint-based representations can additionally prove properties of the encoded rule system (completeness, consistency, unreachability of dead states). The exception-chain collapse pattern documented in §4 is precisely the class of nested structure where this distinction is practically relevant: a three-level exception chain is representable as a decision tree, but verifying that no branch combination produces a contradictory outcome requires the kind of satisfiability check that decision-tree representations do not natively support.

¹The system described in this paper is deployed commercially by Aethis (aethis.ai) for UK immigration and naturalisation workflows. The immigration benchmark sections cover selected requirements from this domain; the full determination involves additional sections not included in this publication.

3.2 LLM Benchmarks for Legal and Logical Reasoning

LegalBench (2023) provides the most comprehensive evaluation of LLM performance on legal tasks, testing 162 tasks across six categories. We report external validation on a subset of LegalBench tasks in §6.10. The two benchmarks remain complementary in scope: LegalBench tests breadth of legal reasoning across 162 tasks; the benchmark in §6.1–§6.9 isolates a single failure pattern under controlled conditions. §6.10 evaluates the architecture on nine LegalBench tasks selected to span the failure pattern (multi-prong rule application) and unbiased-sample tasks (single-clause classification).

FOLIO (2022) tests natural language inference grounded in first-order logic, demonstrating that LLMs struggle with tasks requiring explicit formal logical structure even when relevant premises are provided. Our findings are consistent: the failures we observe are not information retrieval failures (the legislation is provided in full) but reasoning failures arising from the compositional structure of the task.

3.3 Systematic Limits of LLM Reasoning

Dziri et al. (2023) demonstrate fundamental limits of transformer architectures on compositional tasks, showing that performance degrades systematically as task depth increases even when each individual step is within the model’s capability. Exception chain evaluation is compositional in precisely this sense: each individual rule is simple, but the nested application of three independent exception conditions requires compositional reasoning over the rule structure. Our finding that failures concentrate on three-level exception chains and are absent on two-level OR-branching is consistent with the compositional depth hypothesis.

Shi et al. (2023) show that LLMs are systematically distracted by irrelevant context. The related phenomenon of *exemption anchoring* - where attention on the failed primary route suppresses evaluation of valid alternative routes - may be a manifestation of the same attentional bias in a rule-following context.

Valmeekam et al. (2022) demonstrate that LLMs fail reliably on planning tasks requiring state tracking and logical consistency, not through random error but through systematic misapplication of learned heuristics. Our finding that Claude Opus 4.6 produces 0/10 correct responses across ten independent runs on veteran exemption scenarios reflects the same pattern: consistent, confident, and wrong.

Valmeekam, Stechly and Kambhampati (2024) extend this work to reasoning-optimised language models, reporting that OpenAI’s o1 degrades sharply on harder Mystery Blocksworld variants and obfuscated planning tasks — the failure mode is not eliminated by inference-time reasoning compute, merely shifted to deeper compositional depth. We initially read this as parallel to a v3.6 / v3.7 intra-model reasoning-effort finding on the construction-CAR exception chain; that finding was withdrawn in v3.8 (see §6.5 Finding 5) after the original result failed to replicate under instrumented testing. The Valmeekam et al. result remains relevant context for §6.9’s pre-registered N=66 reasoning-effort replication.

Mirzadeh et al. (2024) show that LLM mathematical reasoning is brittle to surface-form changes: renaming entities or adding irrelevant clauses to GSM8K problems systematically degrades accuracy, even on frontier models. This is evidence that what looks like symbolic reasoning is substantially pattern matching over learned surface forms. The exception-chain failure pattern we observe is in the same family — LLMs produce fluent legal-sounding justifications that do not track the actual logical structure of the rule.

3.4 Prompt Engineering and Its Limits

Chain-of-thought prompting (2022) and zero-shot reasoning elicitation (2022) have substantially improved LLM performance on arithmetic and multi-step problems. Wang et al. (2023) show that self-consistency (majority voting) improves performance on reasoning tasks. Our robustness analysis (Section 6.7) explicitly tests whether enhanced prompting closes the accuracy gap on exception chain evaluation, and finds a trade-off rather than a fix: an enhanced prompt that correctly identifies and targets the failure pattern (independent exemption evaluation) reduces false negatives from 7 to 4 while introducing 20 false positives, reducing net accuracy from 90% to 65%. The result demonstrates that prompt-based repair on this class of task is fragile - instructions that correct under-application of exemptions simultaneously cause over-application elsewhere.

3.5 Neuro-Symbolic Architectures and LLM + Formal Method Hybrids

The neuro-symbolic research programme (2009) argues that robust AI systems require integration of neural pattern recognition with symbolic reasoning. Marcus (2020) argues that the reliability limitations of purely statistical systems necessitate a return to hybrid approaches combining learned representations with structured reasoning. Kambhampati et al. (2024) advance this position with the *LLM-Modulo* framework, arguing that LLMs are most robustly deployed as approximate generators paired with formal verifiers and critics that provide external correctness guarantees. The Eligibility Module is a specific instantiation of the LLM-Modulo pattern applied to statutory rule evaluation: LLMs perform pattern-recognition tasks they excel at (reading legislation, extracting structure, generating code), while a constraint evaluation engine handles the evaluation task requiring mathematical guarantees.

Statutory rule evaluation is a particularly favourable application of the LLM-Modulo pattern, in ways that distinguish it from prior instantiations in code synthesis and symbolic planning. Three properties combine. First, the verification fragment is decidable: rule application over compiled constraints terminates in deterministic time with a total correctness criterion, in contrast to test-suite verification of synthesised code (which provides only partial coverage) or constraint satisfaction over learned world models (which is frequently approximate). Second, the artefact being verified — a compiled rule bundle — is persistent and amortised: a single authoring pass serves arbitrarily many subsequent decisions, where per-query LLM-Modulo cycles in code synthesis or planning re-incur authoring cost on every instance. Third, the correctness criterion is externally specified by statute and domain-expert review rather than chosen by the system designer, which makes test-driven validation against expert-defined fixtures (Section 7.3) a meaningful integrity check rather than a tautology. These properties together explain why the LLM-Modulo separation can deliver categorical guarantees at the execution layer in this domain, even where the same separation provides only best-effort guarantees in general program synthesis or planning.

Most directly related is Logic-LM (2023), which uses LLMs to translate natural language problems into formal logical representations, then invokes symbolic solvers for evaluation. LINC (2023) similarly uses LLMs to generate first-order logic programs from natural language for theorem prover evaluation. These systems demonstrate the feasibility of the authoring-execution separation that underlies the Eligibility Module. The present system differs in three respects relevant to high-stakes deployment: it operates on *persistently stored* rule bundles rather than ephemeral per-query translations; it maintains a provenance chain linking each rule to specific source citations; and it is designed for production deployment where audit trails and version control are compliance requirements.

Program-aided language models (PAL (2023)) demonstrate the broader pattern of using LLMs to generate code that is then executed deterministically. The Eligibility Module applies this separation to statutory rule encoding with additional quality engineering (Section 7) to ensure generated rules meet a quality threshold before entering the persistent rule store.

3.6 SMT-Based Constraint Evaluation

Satisfiability Modulo Theories (SMT) solving has seen extensive application in software verification, symbolic execution, hardware design, and safety-critical systems. The Eligibility Module applies SMT-based constraint evaluation to regulatory rule execution - rules compiled from LLM-authored code into formal constraint representations. The near-zero marginal evaluation cost after compilation makes formal constraint evaluation practical for production deployment at scale, a property that distinguishes constraint compilation from per-query LLM inference.

4. The Problem: LLMs and Nested Exception-Chain Evaluation

4.1 What Makes a Decision “High-Stakes”

High-stakes decisions share three properties that distinguish them from general AI tasks:

1. **Material consequence.** An incorrect determination can deny someone a right (citizenship, a benefit, a licence), expose an organisation to regulatory action, or cause financial harm.
2. **Auditability requirement.** A regulator, court, or oversight body may require an explanation of how the determination was reached, traceable to specific rules or statutory provisions.
3. **OR-branching logic.** Rules typically provide multiple independent pathways to satisfaction (primary routes, exemptions, waivers, overrides), each of which is independently sufficient.

Immigration law illustrates this structure. The British Nationality Act 1981 sets out multiple requirements for naturalisation, some of which include alternative pathways to satisfaction — such as age-based exemptions, medical exemptions, and discretionary waivers — which operate as independent routes within those requirements (disjunctive branches), not as standalone routes to overall eligibility (which requires conjunction across requirements).

4.2 Two Failure Patterns

When a large language model is asked to determine eligibility, it processes legislation and applicant data as a single reasoning task. Our benchmark reveals two systematic failure patterns on nested exception-chain tasks in this benchmarked setting. We make no claim that these patterns generalise to all legal reasoning; they are specific to the class of task where exception chains are nested three levels deep:

Failure Pattern 1: Exemption anchoring. LLMs treat exemption and waiver routes as secondary to the primary pathway rather than as independently sufficient alternatives. When the primary route fails, the LLM anchors on that failure and discounts exemptions.

Failure Pattern 2: Exception chain collapse. When rules contain multi-level exception chains (“A is required UNLESS B applies, UNLESS C overrides B”), LLMs fail to correctly evaluate the nested logic.

Table 1: Multi-Level Exception Chain (Spacecraft Benchmark)

The Spacecraft Crew Certification Act (a synthetic statute modelled on UK legislative structure) contains a three-level exception chain for flight readiness:

Level	Rule	Plain English
Base	Flight readiness required (500hrs + licence)	Must have 500+ hours AND a pilot licence
Exception A	Age $\geq$ 60 exempts from flight readiness	Over-60s don’t need flight readiness...
Exception B	...UNLESS mission is orbital	...but orbital missions revoke the age exemption
Override C	1000+ flight hours overrides everything	Veteran pilots (1000+ hrs) are always exempt

Table 2: Failures on Exception Chain Scenarios (spacecraft, adversarial suite)

Scenario	Expected	Opus 4.6	Sonnet 4.6	GPT-5.4	GPT-5-mini
Age 25, 1500hrs, no licence, suborbital	Eligible	1/3	0/3	3/3	1/3
Age 59, 1000hrs, no licence	Eligible	0/3	0/3	3/3	1/3
Age 60, orbital, 999hrs, WITH licence	Eligible	0/3	3/3	3/3	1/3
Age 22, 1001hrs, no licence, orbital + rad cert	Eligible	0/3	3/3	3/3	1/3
Dolphin*, 1200hrs veteran, orbital, provider medical	Eligible	0/3	3/3	3/3	1/3

**Dolphin is a valid species under the synthetic statute (s.3 excludes only Vogons). This scenario tests whether the model correctly applies the veteran exemption to a non-human applicant with otherwise valid credentials.*

The veteran exemption (Override C) is the hardest concept for LLMs in this benchmark. It operates independently of age: a 25-year-old with 1500 flight hours is exempt from flight readiness requirements, just as a 60-year-old would be. Most LLMs tested treat the veteran exemption as age-dependent, producing false negatives with high confidence — Claude Opus 4.6 and Sonnet 4.6 mark the age-59 veteran ineligible on every run (0/3; Table 2), though GPT-5.4 answers the veteran scenarios correctly. The Eligibility Module evaluates these correctly by construction.

4.3 Why False Negatives Matter

In high-stakes decision-making, a false negative means wrongly telling someone they do not qualify. Exemptions exist specifically for edge cases - the people who cannot satisfy the standard route but qualify through an alternative path. These are precisely the applicants most likely to be wrongly denied. On the benchmarked exception-chain scenarios, the failures are systematic: Claude Opus 4.6 returns “ineligible” on every run (0/3) for the age-59 veteran scenario, and on all runs of most other veteran-exemption scenarios in Table 2. Majority voting would not catch this.

Our construction insurance benchmark (Section 6.4) demonstrates the same failure pattern in a different domain: a CAR policy defect exclusion clause with a five-level exception chain. GPT-5.4, which achieves 100% on the spacecraft scenarios, dropped to 96.6% on the insurance scenarios in the March 2026 snapshot (a cell that subsequently closed under model drift; §6.5 Finding 4). The pattern is not specific to a single model, provider, or domain.

5. The Neuro-Symbolic Architecture

5.1 Design Principle

The architecture is built on a simple observation: LLMs and formal constraint evaluation excel at different tasks. LLMs are exceptional at reading legislation, understanding context, and generating structured representations. SMT constraint solvers are exceptional at evaluating logical expressions with mathematical guarantees.

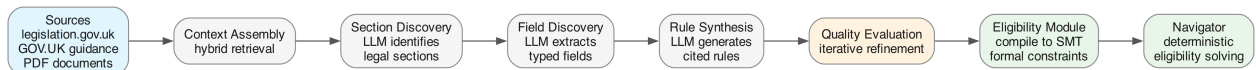
The **Aethis Eligibility Module** separates these concerns:

- **Phase 1: Rule Authoring (LLM).** Read authoritative legal sources, discover sections and fields, generate rules as structured code, evaluate quality iteratively.
- **Phase 2: Rule Execution (Eligibility Module).** Parse generated code, compile to formal SMT constraints, evaluate against applicant data with deterministic execution.

5.2 Phase 1: Automated Rule Authoring

The rule authoring pipeline transforms authoritative legal text into executable compliance rules:

Figure 1: End-to-End Pipeline



Every generated rule carries provenance linking it to specific source material. Source items are tagged with structured references (e.g., [BNA1981#Schedule1/P1.1]) in the LLM’s context, and the LLM is instructed to cite these tags on each generated criterion and field. A verification step then resolves these citations against the source material, creating anchors with document IDs, section paths, and direct quotes:

Table 3: Source Provenance Chain

Source Type	Endpoint	Format	Citation Example	Verification
Primary legislation	legislation.gov.uk	Structured markup	BNA1981#Schedule1/P1.1	Direct resolution from LLM-cited references
Policy guidance	GOV.UK Content API	Structured API	GOVUK#english-lang/part-2	Direct resolution from LLM-cited references
Form guidance	PDF parser	Structured text	form-an#section-4/para-3	Direct resolution from LLM-cited references

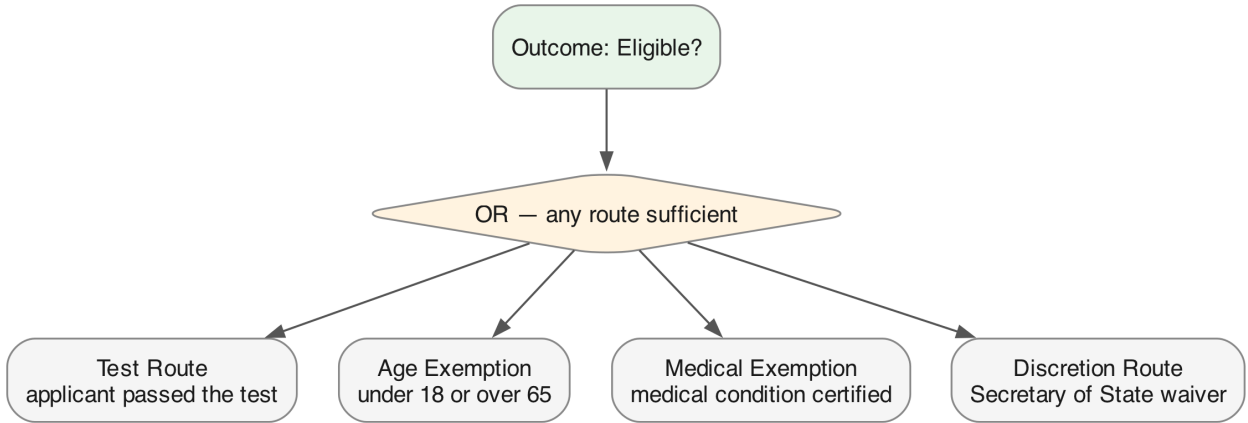
When the LLM fails to cite a source (or cites an invalid reference), the system falls back to embedding-based semantic similarity matching as a secondary mechanism. This multi-stage citation verification with semantic fallback provides higher fidelity than post-hoc similarity matching alone, because the LLM that generated the rule knows which source material it used.

This provenance chain is the foundation of auditability: any determination can be traced through the rule that produced it to the specific source clause that authorises it.

5.3 Phase 2: The Eligibility Module

The authoring model emits rules in a constrained DSL that is parsed and compiled to formal constraints using an SMT-based constraint evaluation engine. The DSL is never executed directly.

The critical property of the compilation is how alternative legal routes are represented. Each eligibility route becomes a formally independent branch:

Figure 2: Route Independence in the Eligibility Module

Each branch is a boolean expression evaluated independently. If `medical_exemption` evaluates to `True`, the OR is satisfied regardless of `test_route`. This is guaranteed by the execution semantics: it is not a statistical property and not dependent on model version or sampling parameters. The guarantee is over execution given the compiled formula; whether the authored formula faithfully represents the source legislation is the Level 2 problem (Section 5.5).

5.4 Why the Eligibility Module Excludes These Errors by Construction

The LLM failures in our benchmark fall into two categories, both of which are excluded by the execution semantics of the Eligibility Module. To make this precise, we state the compilation formally.

Let $\mathcal{F} = \{f_1, \dots, f_m\}$ be the set of typed applicant fields, and let $\sigma : \mathcal{F} \rightarrow \mathcal{V}$ be an applicant assignment (each field mapped to a typed value). A **rule** is a quantifier-free formula ϕ in the theory of bitvectors, linear

integer/real arithmetic, and uninterpreted functions over $\mathcal{F}$. A **rule bundle** is a tuple $\mathcal{R} = \langle \mathcal{G}, \mathcal{O} \rangle$ where $\mathcal{G} = \{G_1, \dots, G_k\}$ is a set of rule groups, each $G_i = \{\phi_{i,1}, \dots, \phi_{i,n_i}\}$ representing independent legal routes within a requirement, and $\mathcal{O}$ is an outcome combinator.

Category 1 — Exemption anchoring (formally excluded). Within a rule group G_i , routes are combined disjunctively:

$$\llbracket G_i \rrbracket_\sigma = \bigvee_{j=1}^{n_i} \llbracket \phi_{i,j} \rrbracket_\sigma$$

The semantics are: $\llbracket G_i \rrbracket_\sigma = \top$ iff *any* route satisfies the applicant assignment. There is no weighting, no ordering, and no context-dependent attenuation of later disjuncts by earlier ones. An LLM’s learned approximation of this disjunction is not guaranteed to exhibit this property; the SMT-evaluated formula is, by the soundness of the decision procedure, evaluated over the authored formula. Whether that authored formula faithfully captures the legislator’s intent is a separate question, addressed at Level 2 (Section 5.5).

Category 2 — Exception chain collapse (formally excluded). A multi-level exception chain of the form

A is required, UNLESS B applies, UNLESS C revokes B, UNLESS D overrides everything

compiles to the material implication

$$\neg \left[\underbrace{\left(\underbrace{(B \wedge \neg C)}_{\text{exemption, not revoked}} \vee \underbrace{D}_{\text{override}} \right)}_{\text{no exemption or override active}} \right] \Rightarrow A$$

or equivalently $(B \wedge \neg C) \vee D \vee A$. This is a closed-form Boolean expression over the applicant fields. Evaluation is a standard SMT decision problem in $O(\text{poly}(|\phi|))$ for the quantifier-free fragment used; the result is a Boolean truth value with no variance over re-evaluation, temperature, or model version.

Contrast with LLM inference. An LLM computes

$$p_\theta(\text{eligible} \mid \text{legislation}, \sigma) \approx \mathbb{1}[\phi_{\text{gt}}(\sigma) = \top]$$

where ϕ_{gt} is the ground-truth formula. The left-hand side is a learned distribution over tokens; the approximation is empirically good on shallow compositional tasks and empirically poor on three-level exception chains (Section 6). The right-hand side — what the Eligibility Module computes — is the indicator function itself. The failure patterns in Section 4 are failures of the approximation, and they are excluded by construction whenever the evaluator operates directly on ϕ_{gt} . The guarantee covers Level 3 (execution) only; whether the authored ϕ faithfully represents ϕ_{gt} is the Level 2 problem (Section 5.5, Section 7).

5.5 What the Eligibility Module Guarantees - and What It Does Not

The system guarantees correctness of execution given a correct specification; it does not guarantee correctness of the specification itself.

Stated more fully: the system provides deterministic guarantees at one level only — **correct execution of formalised rules**. It makes no claim about two upstream steps:

Level 1 - Source text retrieval. The system retrieves authoritative source material using source connectors and retrieval pipelines. Errors here - missing sections, retrieval gaps - will produce incorrectly grounded rules. The system does not guarantee that all relevant source text was found and correctly parsed.

Level 2 - Rule formalisation. LLMs generate rules from source material. An incorrectly formalised rule produces incorrect determinations, even if the execution is formally correct. This is the “garbage in, garbage

out” property stated precisely: deterministic execution of incorrectly formalised rules produces incorrect but deterministic results. The system addresses this through test-driven validation: subject matter experts define test cases covering golden paths, edge cases, and corner cases, and authored rules must pass these test suites before deployment (Section 7.3). Rules can also be reviewed by human experts, but the primary quality gate is automated test-case validation, not manual review.

Level 3 - Rule execution. Once rules are correctly formalised, the Eligibility Module guarantees correct execution with mathematically defined semantics. The failure patterns documented in Section 4 - exemption anchoring and exception chain collapse - are excluded by the execution semantics at this level.

This distinction matters for regulatory defensibility. The provenance chain (Section 9.1) supports audit of Levels 1 and 2; Level 3 is guaranteed by the execution semantics. Execution correctness is a necessary precondition for trust in the overall system: if the execution layer itself could introduce errors, no amount of rule quality engineering would produce reliable determinations. By solving Level 3 first, the system reduces the trust problem to a single surface — rule formalisation quality — which is addressed through iterative synthesis refinement (Section 7.1) and test-driven validation against SME-defined test suites (Section 7.3).

6. Accuracy Benchmark: Eligibility Module vs Eight LLMs

6.1 Methodology

Both systems receive identical inputs: the same legislation text (markdown-formatted) and the same structured applicant data (field key-value pairs). The comparison isolates the reasoning engine. Everything else is held constant.

- **System A (Eligibility Module):** Gold-standard rule fixtures compiled to formal SMT constraints. The navigator evaluates constraints against structured applicant data and returns a deterministic eligible/ineligible determination.
- **System B (LLM baseline):** The same source legislation is provided to each LLM along with the same structured applicant data. The LLM determines eligibility and responds with `{"eligible": true/false, "reasoning": "..."}.`

Models tested:

Model	Provider	Class	Cost Tier
Claude Opus 4.6	Anthropic	Frontier	Premium
Claude Sonnet 4.6	Anthropic	Frontier	Mid
GPT-5.4	OpenAI	Frontier	Premium
GPT-5.3	OpenAI	Production	Mid
GPT-5-mini	OpenAI	Production	Low
GPT-5-nano	OpenAI	Production	Lowest
Claude Opus 4.7	Anthropic	Frontier	Premium
GPT-4.1-mini	OpenAI	Production	Low

Claude Opus 4.7 and GPT-4.1-mini were added in the April 2026 v3.8 replication (Opus 4.7 in Tables 6, 8a, 8c and throughout §6.10; GPT-4.1-mini in Tables 7, 8a, 8b); they are not part of the original March 2026 six-model selection. GPT-4.1-mini serves as a cost-tier baseline for the construction domain.

Evaluation coverage. Not all models were evaluated on all domains. The following matrix shows the scope of each model’s evaluation:

Model	life_uk (56)	english_language (43)	spacecraft (68)	construction_car (58)	Notes
Claude Opus 4.6	✓	✓	✓	✓ (v3.8 replication)	+ robustness (N=10)
Claude Sonnet 4.6	✓	✓	✓	✓ (v3.8 replication)	
GPT-5.4	✓	✓	✓	✓	+ enhanced prompt
GPT-5.3	—	—	—	11-scenario subset	production reference
GPT-5-mini	✓	✓	✓	—	
GPT-5-nano	—	✓	48-scenario baseline	—	cost reference
Claude Opus 4.7	—	—	✓	✓	+ §6.4.1 adversarial; v3.8 replication only
GPT-4.1-mini	—	—	—	✓	construction cost-tier reference

Results reported below should be read against this matrix. Aggregate claims refer only to evaluations actually conducted.

Model selection. The six models span three attributes: (i) frontier versus production tier, (ii) the two providers our organisation had API access to during the evaluation window (Anthropic and OpenAI), and (iii) a range of cost tiers representing realistic deployment choices. Models from other providers (Google Gemini, DeepSeek, Meta Llama, Mistral) were not included in this pass due to access and budget constraints; broader provider coverage is pre-registered for replication at N=66 (Section 6.9). The selection was fixed before any results were computed and was not adjusted based on intermediate findings. Claude Opus 4.7 and GPT-4.1-mini were added later, at v3.8 replication time — Opus 4.7 as the then-strongest Anthropic model and GPT-4.1-mini as a cost-tier reference for the construction domain — which post-dates this original fixed selection.

LLM configuration: Temperature 0 where supported (Claude models; GPT-5 family does not expose this parameter — sampling is controlled internally by the model’s reasoning pipeline). The §6.7 robustness baseline was additionally run at T=0.3 as a temperature-sensitivity control; Table 9 shows the failure set is identical at T=0.3 and T=0. Runs per scenario: 3 (majority vote; robustness analysis tests N=10). Prompt: generic legal assessment, no special instructions about exemptions or OR-branching.

Prompt variants explored. Two to three prompt variants were explored informally during initial development to confirm that the observed failure pattern was not an artefact of wording — for example, whether instructing the LLM to “think step by step” or to “consider all exemption routes before deciding” closed the gap on exception-chain scenarios. These variants did not materially change the pattern and were not logged systematically. Systematic prompt comparison across three variants per scenario is pre-registered for the N=66 replication (Section 6.9). The enhanced prompt analysed in Section 6.7 is a fourth variant specifically targeting the failure pattern.

Benchmark philosophy. The goal is not to find the best possible LLM score through bespoke prompt engineering. It is to test whether a general-purpose model reliably preserves exception structure when reading authoritative rules in the way such systems are actually deployed - with a reasonable, generic instruction. This is a different question from “what is the ceiling of LLM performance on this task?” The robustness analysis (Section 6.7) directly addresses whether the gap is a prompt engineering problem or a deeper reliability issue.

Domain disclosure. The four benchmark domains span different levels of source authenticity:

Domain	Source material	Authenticity
life_uk	British Nationality Act 1981, Schedule 1 + Home Office guidance	Real UK legislation
english_language	Form AN guidance + English language policy	Real UK legislation
spacecraft	Spacecraft Crew Certification Act (synthetic statute)	Synthetic, modelled on UK legislative structure
construction_car	CAR policy defect exclusion endorsement (synthetic wording)	Synthetic, modelled on London market DE3/DE5 clauses

All models evaluated on the `life_uk` section (see the coverage matrix) achieve 100% (depth-1 combinatorial boolean logic); this section is reported here for completeness but not charted separately. All frontier models also achieve 100% on `english_language` (depth-2 multi-route logic); production-tier GPT-5-nano scores 48.8% on the same section (Table 5). These two sections establish a baseline: the failure pattern is specific to exception chain depth, not general to legal reasoning.

Note on immigration sections. The `life_uk` and `english_language` sections cover two specific isolated requirements within UK naturalisation eligibility. The full naturalisation determination involves additional sections with exception chain structures of equivalent or greater complexity to the spacecraft section; those are not included here.

Fixture independence. The rule fixtures used by the Eligibility Module were authored independently of the LLM baseline evaluations and fixed prior to benchmarking. LLMs were not used to generate the ground-truth rules against which they are evaluated.

Scope note on construction insurance materials. The `construction_car` benchmark scenarios and their expected outcomes were authored by the primary author (a non-lawyer) based on published DE3/DE5 clause wordings and industry commentary. They have not been independently validated by a qualified construction-insurance professional and should therefore be read as author-constructed illustrative examples of the exception-chain structure in this domain rather than as professionally validated case studies. The immigration benchmarks (`life_uk` and `english_language`) were authored by L.D., an Immigration Solicitor and accredited Senior Caseworker (see Author Contributions).

Note on repository scope. The public benchmark repository (see Data and Code Availability) additionally contains a `benefits_entitlement` domain that is not analysed in this paper. It is included in the repository for completeness and future work but is outside the scope of this paper; all results, tables and findings reported here cover only the four domains listed above.

Benchmark intent. The benchmark is designed to exercise nested-exception-chain structures that occur in real legislation and commercial contracts, not synthetic structures contrived solely to confuse LLMs. The British Nationality Act 1981 Schedule 1 and JCT 2016 DE3/DE5 insurance wordings both contain multi-level UNLESS chains in practice; we deliberately selected and constructed scenarios that exercise this structure rather than average-case queries against these sources. The finding is therefore about how LLMs handle a structure that recurs in high-stakes legal and insurance texts, not about whether we can synthesise inputs that confuse them. Typical-case performance on the same sources is outside the scope of this paper.

6.2 Scenario Design

Table 4: Benchmark Scenario Coverage

Section	Scenarios	Fields	Difficulty
life_uk	56	4	Low
english_language	43	12	Medium

Section	Scenarios	Fields	Difficulty
spacecraft	68	11	High
construction_car	58	14	High
Total	225		

Key patterns by section. - *life_uk*: combinatorial — 3 booleans $\times$ 7 ages. - *english_language*: Canada trap, SELT edge cases, near-misses, multi-field interactions. - *spacecraft*: three-level exception chain, nested IMPLIES, UNSAT early termination, adversarial. - *construction_car*: DE3/DE5 defect exclusion, access damage carve-back, enhanced cover chain, pioneer override, known defect limitation.

6.3 Results

In all tables below, accuracy figures for the Eligibility Module refer to agreement with the benchmark’s formal rule fixtures — the authored rule specifications used as the benchmark’s ground truth. LLM accuracy is measured against the same fixtures.

Table 5: English Language: 43 Scenarios, 12 Fields (Medium Difficulty)

All accuracy figures report Wilson 95% confidence intervals.

Model	Accuracy	95% Wilson CI	Boundary	Consistency	FN
Eligibility Module	43/43 (100%)	[91.8%, 100%]	7/7 (100%)	43/43 (100%)	0
Claude	43/43 (100%)	[91.8%, 100%]	7/7 (100%)	43/43 (100%)	0
Opus 4.6					
Claude	43/43 (100%)	[91.8%, 100%]	7/7 (100%)	43/43 (100%)	0
Sonnet 4.6					
GPT-5.4	43/43 (100%)	[91.8%, 100%]	7/7 (100%)	43/43 (100%)	0
GPT-5-mini	42/43 (97.7%)	[87.9%, 99.6%]	7/7 (100%)	39/43 (91%)	1
GPT-5-nano	21/43 (48.8%)	[34.6%, 63.2%]	6/7 (86%)	40/43 (93%)	22

Figure 3: Spacecraft accuracy with 95% Wilson confidence intervals.

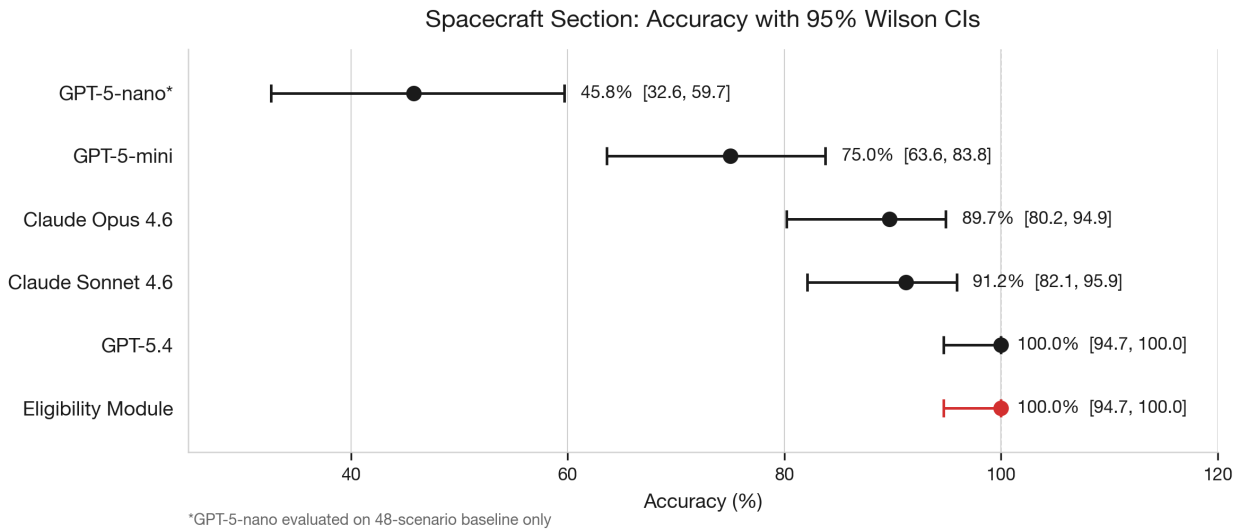


Table 6: Spacecraft: 68 Scenarios, 11 Fields, Adversarial Exception Chains (High Difficulty)

Numbers are the v3.6 / v3.7 snapshot (March 2026); v3.8 replication numbers are listed alongside where they differ. The shifting columns are the central evidence for the model-drift point made in §6.5 Finding 4 and the abstract.

Model	March 2026 (v3.7)	April 2026 (v3.8 replication, same prompt + harness)
Eligibility Module	68/68 (100%) [94.7–100]	68/68 (100%) (deterministic; invariant by construction)
GPT-5.4	68/68 (100%)	not re-tested (was already at 100%)
Claude Opus 4.7 (Anthropic’s strongest at evaluation, Apr 2026)	not in v3.7	68/68 (100%) [94.7–100]
Claude Opus 4.6	61/68 (89.7%) [80.2–94.9]	67/68 (98.5%) [92.1–99.7] — 6 fewer failures
Claude Sonnet 4.6	62/68 (91.2%) [82.1–95.9]	60/68 (88.2%) [78.5–93.9] — 2 more failures
GPT-5-mini	51/68 (75.0%) [63.6–83.8]	not re-tested
GPT-5-nano*	22/48 (45.8%) [32.6–59.7]	not re-tested

**GPT-5-nano was evaluated on the 48-scenario baseline suite only (pre-adversarial expansion) and is included as a cost-scaled reference, not as a realistic deployment choice.*

Replication harness: `tools/replication_run.py` in the public repo, run via the paper’s exact `_build_prompt` format. Per-scenario JSONs at `docs/replication/A*.json`. The Opus 4.6 89.7% → 98.5% drift between March and April invalidates the §6.7 R1 70-trial-on-7-failing-scenarios result as currently written; see §6.7 R1 caveat.

6.4 Construction Insurance: DE3/DE5 Defect Exclusion (58 Scenarios)

The construction insurance benchmark extends the evaluation into a domain with commercial relevance: London market Construction All Risks (CAR) policy wording. The test fixture is modelled on DE3/DE5 defect exclusion clauses used for major infrastructure projects.

The policy logic. A CAR policy covers physical loss or damage to the Works. The defect exclusion creates a five-level exception chain (a base exclusion modified by five successive levels of exception; scenario labels such as “depth-6” count the base as level 1):

Level	Rule	Plain English
Exclusion	Rectification cost absolutely excluded	Cost of fixing the defective part: never covered
Carve-back	Resultant damage to other property covered	Non-defective property damaged by the failure: covered
Re-exclusion	Access damage excluded	Property damaged just to reach the defect: not covered
Enhanced cover	Projects ≥ £100M: access damage reinstated	Large projects get access damage back...
Design limit	...except for design defects	...but not if the defect was in the design
Pioneer override	Projects ≥ £500M: design limit removed	Mega-projects override the design limitation

Table 7: Construction Insurance Exception Chain (access damage scenarios)

Cells reflect the v3.7 snapshot (March 2026). The “Access, £500 M, design” cell where GPT-5.4 was reported as **x** does not replicate in April 2026 — current GPT-5.4 returns the correct answer (covered) on this scenario; see Table 8a v3.8 column. The cell is preserved here as the v3.7 historical record because the model-drift evidence is itself part of the paper’s argument.

Scenario	Project	Defect	Eligibility Module	GPT-5.4 (v3.7)	GPT-4.1-mini
Access, £100M, workmanship	£100M	workmanship	COVERED	✓	x
Access, £200M, materials	£200M	materials	COVERED	✓	x
Access, £200M, design	£200M	design	NOT COVERED	✓	✓
Access, £500M, design	£500M	design	COVERED	x (v3.7) → ✓ (v3.8)	x
Access, £800M, design	£800M	design	COVERED	✓	x

Figure 4: Construction insurance accuracy with 95% Wilson confidence intervals.

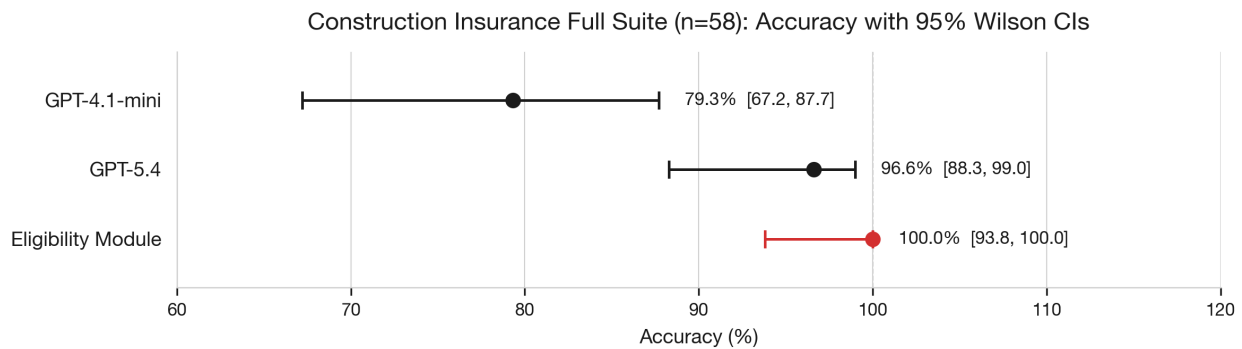


Table 8a: Construction Insurance — Full Suite (March 2026 vs April 2026 replication)

Primary results on the 58-scenario construction insurance benchmark. v3.7 numbers (March 2026 snapshot) shown alongside v3.8 replication (April 2026, same prompt and harness, current model alias). Wilson 95% CIs in brackets.

Model	March 2026 (v3.7)	April 2026 (v3.8 replication, full 58-suite + 16 early scenarios from the §6.9 exception-chain expansion set)
Eligibility Module	58/58 (100%) [93.8–100.0]	74/74 (100%) [95.1–100] (deterministic; invariant by construction)
GPT-5.4	56/58 (96.6%) [88.3–99.0]	74/74 (100%) [95.1–100] — closed silently between March and April
Claude Opus 4.7 (Anthropic’s strongest at evaluation, Apr 2026)	not tested in v3.7	74/74 (100%) [95.1–100]
Claude Opus 4.6	not tested in v3.7	74/74 (100%) [95.1–100]

Model	March 2026 (v3.7)	April 2026 (v3.8 replication, full 58-suite + 16 early scenarios from the §6.9 exception-chain expansion set)
Claude Sonnet 4.6	not tested in v3.7	72/74 (97.3%) [90.7–99.3] — fails on access_100m_design_not_covered, surface_eligible_design_access_blo
GPT-4.1-mini	46/58 (79.3%) [67.2–87.7]	65/74 (87.8%) [78.5–93.5] — direction holds; magnitude shifted; still fails on neg_* (negation_stacking), surface_* (contradictory_cue), and ultimate_trap_1

GPT-5.3 was the cost-tier production reference in v3.7 (claimed 7/11 on the n=11 exception-chain subset). The model alias was deprecated by OpenAI between March and April 2026 (NotFoundError: model gpt-5.3 does not exist). Replication impossible.

The v3.7 paper’s lead frontier-LLM failure case (GPT-5.4 missing 2/58 construction scenarios) does not replicate in v3.8. The v3.4 / v3.7 narrative “*no frontier model achieves 100% across all four paper domains*” must be qualified: as of April 2026, GPT-5.4, Opus 4.7, and Opus 4.6 all reach 100% on the v3.7 58-scenario construction benchmark using the same paper-prompt format. Sonnet 4.6 still fails 2/74; GPT-4.1-mini still fails 9/74; the cost-tier and a small set of mid-tier failure modes remain visible. **The v3.7 paper-suite no longer differentiates the engine from the strongest frontier configurations on the controlled benchmark — and that is itself the point.** §6.4.1 below documents the v3.8 adversarial extension that does still differentiate, demonstrating that the structural advantage holds at deeper composition.

Replication artefacts: `confidently-wrong-benchmark/legalbench/docs/replication/A*.json` and `STREAM_A_REPORT.md` for full per-scenario JSON, including which specific scenarios each model failed on.

Table 8b: Construction Insurance — Exception-Chain Sub-Study (N=11)

Inter-model comparison on the 11-scenario `exception_chain`-tagged subset. Wilson 95% CIs in brackets. The original v3.6 / v3.7 row “GPT-5.4 (low reasoning) — 7/11 (63.6%)” is **withdrawn in v3.8** following an instrumented replication that returned 11/11 (100%) correct under explicit `reasoning_effort=low`; see `docs/r5-withdrawal-note.md` and Finding 5 above. The §6.9 pre-registered N=66 replication remains the venue for any larger-sample reasoning-effort test.

Model	Exception-chain (n=11)	Notes
Eligibility Module	11/11 (100%) [74.1–100.0]	deterministic by construction
GPT-5.4 (default)	10/11 (90.9%) [62.3–98.4]	original v3.6 / v3.7 figure
GPT-5.4 (<code>reasoning_effort=low</code> , v3.8 replication)	11/11 (100%) [74.1–100.0]	instrumented; cannot reproduce v3.6 / v3.7 7/11
GPT-5.3	7/11 (63.6%) [35.4–84.8]	production-tier baseline
GPT-4.1-mini	5/11 (45.5%) [21.3–72.0]	cost-tier baseline

GPT-5.3 scores 7/11 (63.6%, 95% CI [35.4%, 84.8%]) on the 11-scenario exception chain subset. It correctly identifies absolute exclusions and simple carve-backs but fails on four scenarios requiring

multi-level exception reasoning, including enhanced cover reinstatement, the pioneer override, and the depth-5 unblock. The v3.6 / v3.7 paper additionally claimed GPT-5.4 at `reasoning_effort=low` matched this 7/11 with the same four scenarios failing — that intra-model reasoning-effort claim is **withdrawn in v3.8**: an instrumented replication on the exact same 11 scenarios returned 11/11 correct under explicit `reasoning_effort=low`, and no committed script reproduces the original 7/11 figure (see Finding 5 above and `docs/r5-withdrawal-note.md`). The inter-model gap (GPT-5.3 64% vs GPT-5.4 default 91% on the same chain) is real and narrow; §6.9 pre-registers an N=66 replication for the deeper compute-dependence question.

Note: v3.5 of this paper reported GPT-5.3 at 27% (3/11). That figure was inflated by a harness configuration issue in which the output budget was too small for reasoning models — their internal chain-of-thought tokens consumed the budget before visible output was produced, which scored as errors. With the corrected budget, GPT-5.3’s actual accuracy is 64%. The correction is detailed in Section 6.8.

GPT-5.4 was reported in March 2026 to fail on the pioneer override boundary: `access_500m_design` was incorrectly rejected (the override applies at $\geq$ £500 M), while the identical scenario at £800 M was correctly accepted. **In April 2026 this specific cell no longer reproduces**: GPT-5.4 returns the correct verdict (covered) on `access_500m_design` under the same prompt. GPT-4.1-mini’s enhanced-cover-chain failure is still present in v3.8 replication. The Eligibility Module evaluates `500 >= 500 = True` with no ambiguity, regardless of model drift.

6.4.1 v3.8 Adversarial Extension (20 scenarios)

The v3.7 58-scenario suite no longer reliably differentiates the strongest frontier configurations from the Eligibility Module on the controlled benchmark — the specific failure cells documented in March 2026 have largely closed at current model performance. To re-establish a current frontier-LLM failure demonstration on the same domain (and to test whether the structural advantage of deterministic execution is real or transient), we authored 20 new construction-CAR scenarios stratified across five complexity dimensions:

- **A — Maximum-stack adversarial**: three or more rule-failure modes simultaneously (e.g. depth-6 pioneer override + non-JCT existing-structures + design defect + access damage).
- **B — Threshold-boundary edge cases**: at-threshold (£100 M, £500 M), just-below (£99 M, £499 M), just-above (£100 M+ ϵ , £500 M+ ϵ).
- **C — Multi-clause cross-reference**: existing-structures + JCT + design + pioneer interactions across Cl. 5, 9, 10.
- **D — Surface-vs-deep contradiction**: surface text suggests opposite of correct answer (e.g. “£800 M project, design defect, access damage” surface signal = not covered; correct answer = covered via pioneer override).
- **E — Conjunction tracking**: every condition near-violating; one boolean flip reverses the outcome.

By design, these 20 scenarios stress-test specific compositional failure modes. They are not claimed to be representative of typical legislative drafting frequency or distribution; they are an adversarial probe at the upper end of the complexity stack the Module is intended to handle, included to demonstrate that the structural advantage persists when frontier models are pushed beyond the v3.7 baseline difficulty.

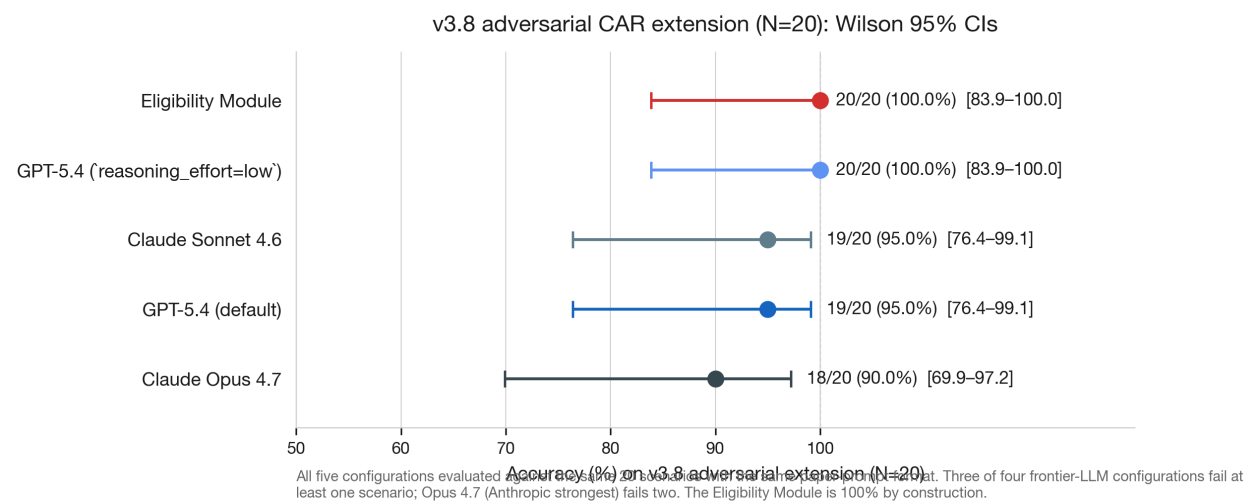
Methodology — addressing the code-derived ground-truth critique. Each scenario was authored with an *independent prose justification* for its expected outcome before the engine was consulted. The prose lives in the scenario’s notes: field; the binary expected-value (`expect.outcome`) was authored in the same pass for consistency. The scenarios were then submitted to the engine via the public `/api/v1/public/decide` endpoint against the canonical bundle `construction-all-risks:20260412-gold` (the same bundle the full 74-scenario replication suite of Table 8a uses; the rule encoding was *not* modified for the v3.8 extension). Engine outcome matched authored expectation on 20/20, indicating self-consistency between the prose, the binary, and the rule encoding. The frontier-LLM runs (below) provide a third independent check: when models read the same `source.md` and answer the same scenarios, their disagreements with engine reveal model failures, not authoring errors (see §6.4.1 cross-evaluation note).

Frontier-LLM evaluation. The 20 new scenarios were run against four frontier-LLM configurations using the paper’s exact `_build_prompt` format from `confidently-wrong-benchmark/benchmarks/run_llm_comparison`. All five evaluations (engine + 4 LLM configs) ran independently against the same scenario inputs.

Table 8c: v3.8 Adversarial Extension — 20 scenarios, 4 frontier-LLM configurations

Configuration	Correct	Wilson 95% CI	Failure scenarios
Eligibility Module	20/20 (100%)	[83.9–100]	—
GPT-5.4	20/20 (100%)	[83.9–100]	—
(reasoning_effort=low)			
Claude Sonnet 4.6	19/20 (95.0%)	[76.4–99.1]	v38_e4_carveback_gap_explic
GPT-5.4 (default)	19/20 (95.0%)	[76.4–99.1]	v38_e4_carveback_gap_explic
			0 reasoning tokens
			on every scenario
Claude Opus 4.7	18/20 (90.0%)	[69.9–97.2]	v38_b3_499m_workmanship_acc
(Anthropic’s			v38_e4_carveback_gap_explic
strongest at			
evaluation, Apr			
2026)			

Figure 9: v3.8 adversarial extension forest plot.



Cross-evaluation note: the carveback-gap scenario (E4) is failed by three of four frontier-LLM configurations across both Anthropic and OpenAI families. The scenario tests the DE3/LEG3 coverage gap explicitly documented in `source.md` Cl. 7 commentary: with `is_access_damage=true` and `consequence_of_failure=false`, the carveback group fails (Route A needs consequence; Route B needs not-access). The £200 M project value qualifies for enhanced cover, but the prior carveback gate blocks the claim before the enhanced-cover logic is reached. GPT-5.4 default and Sonnet 4.6 return `eligible`; Opus 4.7 returns `eligible`; correct answer is `not_eligible`. Only GPT-5.4 at `reasoning_effort=low` — the configuration that forces the model to use reasoning tokens (95 tokens on E4) rather than short-circuit (default = 0 reasoning tokens on every scenario) — gets it right. The pattern is consistent with a structural compositional-evaluation failure mode rather than a per-model artefact.

Reasoning-effort observation. GPT-5.4 at default reasoning effort uses **0 reasoning tokens on 20 of 20 v3.8 adversarial scenarios** — the API returns the answer in 4–6 completion tokens with no chain-of-thought. At `reasoning_effort=low`, the same model uses 16–126 reasoning tokens per scenario. On the harder cases this additional deliberation matters: low-reasoning gets E4 right where default short-circuits to wrong. The v3.6 / v3.7 paper claim that “default reasoning is better than low reasoning” (Finding 5, withdrawn)

appears to be inverted under current model behaviour — default short-circuits without reasoning; low forces deliberation.

Replication artefacts: `tools/replication_run.py`, `tools/run_engine_v3_8_adversarial.py`, scenario YAML at `dataset/construction-all-risks/scenarios_v3_8_adversarial.yaml`, per-scenario JSONs at `docs/replication/B3_*.json` and `docs/replication/B4_*.json`, full report at `docs/replication/STREAM_B_REPORT.md`. All in the public `confidently-wrong-benchmark/legalbench/` (paths relative to that repo root).

6.5 Error Analysis

Six findings emerge from the multi-model benchmark (one, Finding 5, was withdrawn in v3.8 and is retained below for the record):

Finding 1: The failure pattern is exception-chain specific. All frontier models achieve 100% on `english_language` (43 scenarios with multi-route logic). On the spacecraft section with its three-level exception chain, Claude Opus 4.6 drops to 90% and GPT-5-mini to 75%. The breaking point is nested exceptions, not legal reasoning in general.

Finding 2: The baseline failures are false negatives. Across the March 2026 baseline evaluations (§6.3–§6.4), we observed no false positives: no LLM ever declared an ineligible applicant eligible or a non-covered claim covered. The baseline failure mode is exclusively conservative - but in a high-stakes context, a false negative may deny someone a pathway the rules explicitly provide. Two later evaluation arms are explicit exceptions to the no-false-positive observation: the §6.4.1 v3.8 adversarial extension, where three frontier configurations (GPT-5.4 default, Sonnet 4.6, and Opus 4.7) return `eligible` on the `not_eligible` E4 carveback-gap scenario — false positives (§6.4.1); and the §6.10 LegalBench classification baselines, which exhibit positive-class anchoring under the upstream few-shot prompt format, also producing false positives (see §6.10.4).

Finding 3: The veteran exemption is the universal failure point. The most-failed scenario across all models is `age_59_veteran_1000hrs`. Claude Opus 4.6 and Sonnet 4.6 mark this person ineligible on every run (0/3); GPT-5-mini answers correctly on only 1 of 3 runs; GPT-5.4 alone is correct on all runs (Table 2). The LLM conflates two independent exemption pathways.

Finding 4 (revised v3.8): Frontier model accuracy is unstable across model updates and prompt configurations. The v3.6 / v3.7 paper reported that GPT-5.4 dropped to 96.6% on the construction insurance section while achieving 100% on spacecraft and immigration. **The v3.8 replication finds that specific failure case has closed:** under the same paper-prompt format and harness, current `gpt-5.4` is 74/74 (100%) on the full construction-CAR suite (Table 8a). The Opus 4.6 spacecraft accuracy also rose from 89.7% (March) to 98.5% (April) under the same `claude-opus-4-6` alias. No model-version bump was announced; the specific empirical failures changed silently between months. The structural failure pattern remains observable on the v3.8 adversarial extension (§6.4.1: Opus 4.7 fails 2/20, GPT-5.4 default fails 1/20, Sonnet 4.6 fails 1/20), and on §6.10 LegalBench (combined McNemar’s $p < 0.001$ vs each frontier model). For a regulated workflow built on a benchmark-time accuracy claim, the central practical observation is that the specific model performance is a moving target — even within a single model alias — while the deterministic execution layer is invariant by construction.

Finding 5 — WITHDRAWN in v3.8. Earlier revisions of this paper (v3.6, v3.7) reported, as a tentative N=11 finding, that GPT-5.4 at `reasoning_effort=low` scored 7/11 (63.6%) on the construction-CAR exception-chain subset, matching GPT-5.3 at default reasoning. **The v3.8 polish pass could not reproduce this result.** Two converging issues: (a) the committed LLM-comparison harness in `confidently-wrong-benchmark/benchmarks/run_llm_comparison.py` does not pass `reasoning_effort` as an OpenAI API parameter, and no other committed script produced the v3.6 / v3.7 figure — the original result has no committed-code provenance; (b) an instrumented v3.8 replication on the exact 11 `exception_chain`-tagged scenarios with explicit `reasoning_effort=low` and full per-call logging returns 11/11 (100%) correct, with all responses non-empty (`finish_reason=stop`) and

reasoning-token counts in the expected 13–166 range. The parsimonious explanation is that the v3.6 / v3.7 result was a harness artefact analogous to the v3.5 → v3.6 GPT-5.3 token-budget bug (§6.8 *Harness configuration correction*); we cannot rule out model drift since the original test, but the absence of a reproducible script is on its own sufficient to retract. Full withdrawal note, replication command, and per-scenario JSON: docs/r5-withdrawal-note.md and docs/verify_gpt5_reasoning_n11.json. Section 6.9’s pre-registered N=66 replication is retained as the venue in which any larger-sample reasoning-effort effect would be detected.

Finding 6 (revised v3.8): The shifting-ground problem. Multiple v3.6 / v3.7 frontier-LLM cells do not replicate under the same prompt and harness six weeks later. The most concrete examples: GPT-5.4 on construction-CAR moved from 96.6% to 100% (Table 8a); Opus 4.6 on spacecraft moved from 89.7% to 98.5% (Table 6); the v3.6 / v3.7 GPT-5.4 reasoning-effort-dependence claim (Finding 5) was withdrawn after an instrumented replication produced the opposite result; the GPT-5.3 model alias was deprecated by OpenAI mid-paper-cycle. None of these are catastrophic individually, but the pattern is that *frontier-LLM accuracy on a fixed benchmark is a function of the model snapshot, the harness configuration, and the prompt format — and at least one of these can shift without notice*. For a regulated workflow that depends on benchmark-time accuracy claims to certify deployment, this property is structurally incompatible with the verification pipelines required by frameworks like the EU AI Act. The Eligibility Module avoids this class of risk by construction: rules are compiled once at authoring time and evaluated deterministically thereafter; the same bundle gives the same answer on the same inputs in March, April, or any subsequent month, regardless of any silent change to upstream model behaviour. The v3.8 adversarial extension (§6.4.1) is published precisely so that this paper’s frontier-LLM numbers are themselves replicable from the committed harness — not as a snapshot but as a procedure.

6.6 The Cost-Accuracy Trade-off

Property	Eligibility Module	GPT-5.4 (best LLM)
Deterministic	Yes (by construction)	No
Cost per evaluation	Near-zero marginal after compilation	~\$0.02 per scenario
Latency	<1ms	~2–5 seconds
Auditability	Exact logical path	Post-hoc rationalisation
Regulatory defensibility	Execution semantics defined	Statistical accuracy
Model deprecation risk	None	Version-dependent

A production system cannot guarantee that GPT-5.4’s 100% accuracy on this benchmark extends to all possible inputs. It is an empirical observation, not an execution-level guarantee.

6.7 Robustness Analysis

This section addresses the key question: whether the observed failures are prompt-sensitive or structurally robust. The central question is not “what is the best possible LLM score after bespoke optimisation?” but “when models fail on exception chains, is that a prompting artefact - fixable with better instructions - or a deeper reliability issue?”

Table 9: Robustness Analysis (spacecraft section, 68 scenarios)

Condition	Claude Opus 4.6	GPT-5.4
Generic prompt, T=0.3 (Claude) / API default (GPT), N=3 (<i>baseline</i>)	90% (7 FN)	100%
Generic prompt, T=0, N=3 (<i>Claude only</i>)	90% (7 FN, all 0/3)	N/A*

Condition	Claude Opus 4.6	GPT-5.4
Generic prompt, T=0.3, N=10 (<i>Claude only</i>)	90% (7 FN, all 0/10)	-
Enhanced prompt, N=3	65% (4 FN + 20 FP)	99% (1 FN)

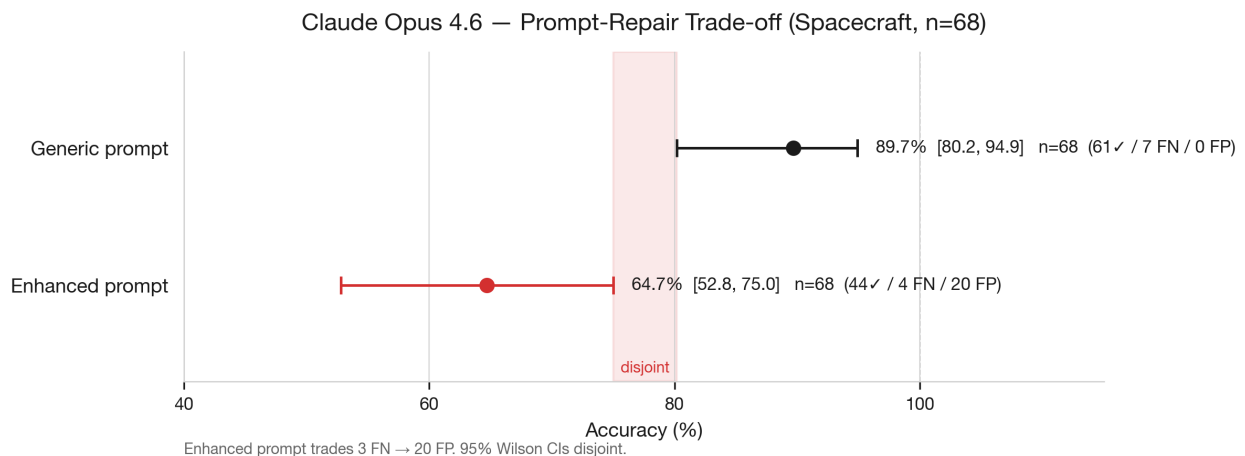
**GPT-5 family does not expose a temperature parameter.*

R1 (March 2026 snapshot; partial v3.8 caveat). At temperature 0 (fully deterministic), Claude Opus produced the same 7 failures with the same 0/3 error rate at the time of the original test. With 10 runs per scenario, all 7 failed scenarios scored **0/10**. Aggregated across the 7 scenarios, this is **0 correct answers in 70 independent trials**. The Clopper–Pearson 95% one-sided upper bound on per-trial success probability is **4.19%**, ruling out the hypothesis that the model would produce correct answers at a rate $\geq 5\%$ per attempt at that snapshot.

v3.8 caveat. The “7 failing spacecraft scenarios” baseline that R1 was tested on (Opus 4.6 = 61/68 in March) does not replicate today: in April 2026 Opus 4.6 returns the correct answer on 67 of 68 spacecraft scenarios under the same prompt and harness, leaving only 1 systematic failure rather than 7 (Table 6 v3.8 column). The 70-trial 0/70 result therefore stands as a March 2026 finding about a specific 7-scenario set that is no longer at the same baseline failure rate. Re-running the 70-trial design against the current 1-scenario failure set would require a separate investigation we have not undertaken in v3.8. The headline qualitative claim — *that some specific Opus failures were stable rather than stochastic at the original snapshot* — survives. The headline quantitative claim — *that 4.19% upper bound applies to current Opus 4.6 production behaviour on current failure cases* — does not.

R2: Enhanced prompting trades one failure mode for another. The enhanced prompt explicitly instructs the LLM to “evaluate each exemption independently” and warns about exception chains. This partially succeeds: the false-negative count falls from 7 to 4 (with run-to-run variance in which specific scenarios fail; see R4). However, the same instructions cause the model to over-apply exemption logic, producing **20 false positives** — the first time in our benchmark that an LLM declares an ineligible applicant eligible. The model now grants eligibility to applicants below the flight hours threshold ($499 < 500$), with expired medical certificates, and on orbital missions where the age exemption is explicitly revoked. Net accuracy drops from 61/68 (89.7%, CI [80.2%, 94.9%]) to 44/68 (64.7%, CI [52.8%, 75.0%]) — the CIs are disjoint, so the drop is statistically significant.

Figure 5: Prompt-repair trade-off (Claude Opus 4.6, spacecraft).



We note that R2 tests only one enhanced prompt variant. Section 6.9 pre-registers a three-variant comparison (few-shot, self-critique, decomposition) to test whether this FN/FP trade-off is fundamental to prompt repair on this class of task or specific to the particular prompt tested here.

This is not simply a bad prompt. The enhanced prompt correctly identifies the reasoning gap (independent

exemption evaluation) and provides accurate instructions. The problem is that the same instructions that fix under-application of exemptions cause over-application elsewhere. The generic prompt’s conservative bias - all false negatives, no false positives - is a more predictable failure mode than the enhanced prompt’s scattered errors across both directions. For a production system, a consistent conservative bias is easier to compensate for than unpredictable failures in both directions.

R3: GPT-5.4 shows the same pattern in miniature. The enhanced prompt introduces 1 failure on the hardest scenario that the generic prompt handles correctly.

R4: Run-to-run variance confirms fragility. Re-running the enhanced prompt benchmark on the same model produces qualitatively identical results (20 FP, 4 FN) but with slight variance in which specific scenarios fail. This contrasts with the generic prompt, where the same 7 scenarios fail identically across independent runs. The enhanced prompt introduces not only new failure modes but also non-deterministic failure patterns - precisely the property a high-stakes system cannot tolerate.

R5 — WITHDRAWN in v3.8. Earlier revisions of this paper reported that GPT-5.4 at `reasoning_effort=low` scored 7/11 on the construction-CAR exception chain, suggesting reasoning-compute-dependence within the GPT-5 family. The v3.8 polish pass found (a) no committed script reproduces the v3.6 / v3.7 result — the LLM-comparison harness at `confidently-wrong-benchmark/benchmarks/run_llm_comparison.py` does not pass `reasoning_effort` as an API parameter, and (b) an instrumented replication on the exact 11 `exception_chain`-tagged scenarios with explicit `reasoning_effort=low` returns 11/11 (100%) correct, with all responses non-empty and reasoning tokens in the 13–166 range. Most likely cause: the v3.6 / v3.7 figure was a harness artefact analogous to the v3.5 → v3.6 GPT-5.3 token-budget bug. The intra-model reasoning-compute-dependence hypothesis is therefore **withdrawn**. Claude Opus 4.6’s temperature invariance (100% at both default T and T=1.0) stands as a separate observation but no longer serves as a parallel data point to a withdrawn claim. Section 6.9’s pre-registered N=66 replication is retained as the venue for any larger-sample test of reasoning-effort dependence. Withdrawal note + per-scenario JSON: `docs/r5-withdrawal-note.md`, `docs/verify_gpt5_reasoning_n11.json`.

The robustness findings address the prompting-contest objection at the level of the strategies tested here. The gap is not caused by a suboptimal generic prompt that the enhanced variant fixes. On the two prompts tested, prompt engineering on this class of task exhibits a trade-off: instructions that reduce false negatives on exception chains simultaneously increase false positives by causing over-application of the same exemption logic. This is consistent with the compositional reasoning limitation identified by Dziri et al. (2023): the model does not have a stable internal representation of the exception chain structure that can be steered by prompt instructions without side effects. We find no evidence that the prompting strategies tested here reliably resolve the failure mode; whether richer strategies (few-shot, self-critique, decomposition, tool-augmented) close the gap is the pre-registered subject of §6.9.

6.8 Threats to Validity

We follow academic convention in identifying threats to the validity of these findings.

Internal validity: - *Prompt sensitivity.* The benchmark uses a generic prompt, not an optimised one. This is intentional (Section 6.1), but means we are not measuring best-possible LLM performance. The robustness analysis (Section 6.7) tests one enhanced prompt variant that specifically targets the identified failure pattern; it reduces false negatives but introduces false positives, with net accuracy decreasing. We cannot exclude the possibility that a different prompt strategy could improve accuracy without side effects, but the result demonstrates that targeted prompt repair on this class of task involves trade-offs between failure modes. - *Harness configuration correction (v3.6).* v3.5 reported GPT-5.3 at 27% (3/11) on the construction exception chain. That figure was inflated by a configuration issue in the benchmark harness: the output budget was too small for reasoning models, whose internal chain-of-thought tokens consumed the budget before visible output was produced, resulting in empty responses scored as errors. With the corrected budget, GPT-5.3’s actual accuracy is 64% (7/11). The correction does not affect the qualitative finding — GPT-5.3 still fails systematically on multi-level exception scenarios — but the quantitative severity is less extreme than origi-

nally reported. The benchmark script has been corrected and the updated results are publicly reproducible. - *Expected values partly code-derived.* The `life_uk` expected values are computed by Python code from the section’s authored eligibility formula (Appendix A.1); L.D. authored and validated the legal formalisation (Author Contributions), but the 56 generated expected values were not individually hand-verified. The `english_language` scenarios were hand-verified against Form AN guidance. The spacecraft scenarios use a synthetic statute with unambiguous values. - *Structured inputs only.* Applicant data is provided as typed key-value pairs. A production system requires extraction from natural language or documents; that layer is not tested here. - *Adversarial scenarios are post-hoc.* The spacecraft adversarial suite was designed after observing LLM failure patterns. The baseline results (48 scenarios) show the same failure pattern; the adversarial additions amplify it. Selection bias of this kind is addressed for the LegalBench external-validation arm by §6.10.3’s pre-registered random-sample replication (seeds 42, 43, 44).

External validity: - *Synthetic domains.* The spacecraft domain uses a synthetic statute. The construction insurance domain uses synthetic policy wording modelled on real clauses. Findings in these domains may not generalise directly to all legal contexts. §6.10 provides cross-domain external validity on a public peer-reviewed legal-reasoning benchmark (LegalBench). - *Benchmark scope.* The benchmark tests one specific class of task — nested exception-chain evaluation. LLM performance on other legal reasoning tasks may differ substantially; §6.10 reports both the multi-prong (failure-pattern-matching) and single-clause-classification (random-sample) regimes. - *LegalBench prompt format.* The low GPT-5.4 cells in §6.10.4 are tied to the upstream LegalBench few-shot classification prompt format. We report them because they affect the canonical baseline under the evaluated protocol, but they should not be read as best-achievable GPT-5.4 performance estimates. A broader prompt sweep would be required before treating those cells as model-capability estimates. - *Model versioning.* Specific model versions are named. As models are updated, results may change. The benchmark scenarios are released publicly to allow ongoing evaluation of future systems. The specific accuracy figures reported here are a snapshot; the benchmark task structure and failure pattern are the durable contribution.

Construct validity: - *LLM self-evaluation in quality engineering.* Quality scores in Section 7 are produced by an LLM evaluator, not human legal experts. These are internal consistency scores; external validation has not been performed.

6.9 Pre-Registered Replication (v3.7)

The prompt-repair finding (R2) rests on one enhanced prompt variant; the reasoning-effort question (formerly Finding 5 / R5, withdrawn in v3.8 — see §6.5) is now an open hypothesis rather than a tentative finding. v3.7 of this paper pre-registers the following replication, which v3.8 retains as the venue for both questions:

1. **Expanded construction_car exception-chain subset from 11 to 66 scenarios**, stratified across five depth bands (simple exclusion, exclusion+carve-back, +re-exclusion, +enhanced cover, +design-limit/pioneer/depth-5) with boundary and adversarial coverage within each band. Power analysis: N=35 per arm is required to detect a 64% vs 91% effect at 80% power, $\alpha=0.05$; N=66 gives $\sim 1.9\times$ margin and tightens the Wilson 95% CI at 64% observed accuracy from $\pm 25\text{pp}$ (at N=11) to $\pm 11\text{pp}$.
2. **Three prompt-variant robustness tests** on the 7 known-failing spacecraft scenarios plus 3 control scenarios (N=3 runs each), across Claude Opus 4.6 and GPT-5.4: (a) few-shot with worked exception-chain examples, (b) self-critique two-stage, (c) decomposition (evaluate each route separately, then OR). Primary outcome: does any variant escape the FN/FP trade-off observed in R2?
3. **Full reasoning-effort sweep** on the 66-scenario exception chain: GPT-5.4 at low/medium/high, GPT-5.3 at default (and at low/high if the parameter is exposed), Claude Opus 4.6 as invariance control.
4. **Independent expected-value authoring for new scenarios:** each of the 55 new scenarios has a prose expected-value assertion written against the legislative clause chain before the fixture is consulted. Mismatches between prose and fixture are flagged for SME review rather than silently reconciled. This addresses the internal-validity hole where code-derived expected values allow fixtures and the execution engine to trivially agree.

Pre-registration commitment. Results — whether they confirm, weaken, or refute the earlier findings — will be reported in a subsequent revision regardless of direction. (Status as of v3.11, July 2026: 16 of the 55 pre-registered new exception-chain scenarios were authored and exercised in the v3.8 construction-CAR replication — they form the 58→74 expansion behind Table 8a — but the full N=66 sub-study, with its three prompt-variant arms and reasoning-effort sweep, remains pre-registered and not yet run; it is the next experimental milestone for this paper.) Findings 5 and 6, and Section 6.7 R5, will be updated or withdrawn based on the replication.

6.10 External Validation on LegalBench

The benchmark in §6.1–§6.8 isolates a single failure pattern (nested exception-chain evaluation) on synthetic and UK-immigration scenarios authored for the experiment. A natural external test of the architecture is a public, peer-reviewed legal-reasoning benchmark covering tasks the authors did not design. We take that test using LegalBench (2023), evaluating the Eligibility Module on nine LegalBench tasks (949 held-out cases) against three frontier LLMs spanning two model families.

6.10.1 Setup

For each LegalBench task we evaluate, we author one rule bundle for the Eligibility Module from the **verbatim canonical Task description** published in the upstream LegalBench repository (CC BY 4.0, Neel Guha). No paraphrase, no editorial shaping, no DSL pseudo-code. The bundle is generated by aethis-core’s authoring pipeline against a small set of statute-derived test cases written by the authors (typically 4–8 per task) — *not* drawn from the LegalBench fact patterns. Where domain-specific subject-matter-expert guidance is added, it lives in a separate guidance/ directory and is grounded in independent practitioner authority (e.g. LSTA Model Credit Agreement conventions for `cuad_covenant_not_to_sue`; standard federal civil-procedure treatments for `personal_jurisdiction`; the Atticus Project’s CUAD taxonomy for clause-classification tasks).

We compare the **Eligibility Module pipeline** (LLM extractor producing typed atomic field values → deterministic engine evaluation against the bundle’s compiled rule) against the **LLM-only baseline** (the same model given the canonical rule prose and asked to answer the LegalBench question end-to-end, using upstream LegalBench prompts — the canonical 5–6-shot `base_prompt.txt` files). LLM baselines are evaluated at the model level for `claude-sonnet-4-6`, `claude-opus-4-7`, and `gpt-5.4`. The Eligibility Module’s runtime extractor in the headline configuration is `claude-sonnet-4-6`; we also report a cross-extractor configuration with `claude-opus-4-7` to disentangle structural from model-specific effects (§6.10.5).

For tasks selected by inspection (matching the failure pattern identified in §4.2), we apply the held-out methodology *retroactively*: results below are filtered to the held-out half of each task’s LegalBench test split using a deterministic seeded partition (`tools/test_split.py --seed 7 --dev_fraction 0.5`). For tasks selected by **pre-registered random sample** from the 153-task LegalBench candidate pool — the 162 upstream tasks minus the nine already integrated at pick time (`tools/random_task_pick.py --seed 42` and `--seed 43`, committed before authoring), the held-out methodology was applied *prospectively* — hint iteration was confined to the dev half, and the holdout was evaluated once with frozen hints.

6.10.2 Results

Table 6.10.A reports per-task accuracy on the held-out evaluation sample, with 95% Wilson confidence intervals on each estimate and exact two-sided McNemar’s tests on the per-case engine-vs-LLM agreement matrices.

Table 6.10.A — Eligibility Module vs frontier LLMs, LegalBench held-out evaluation

Task	N	Eligibility Module	Sonnet 4.6	Opus 4.7	GPT-5.4
hearsay (FRE 801)	47	45/47 (95.7%) [85.8–98.8]	35/47 (74.5%); +21.3 / 0.006	39/47 (83.0%); +12.8 / 0.031	40/47 (85.1%); +10.6 / 0.125
personal_jurisdiction	25	24/25 (96.0%) [80.5–99.3]	24/25 (96.0%); +0.0 / 1.000	23/25 (92.0%); +4.0 / 1.000	24/25 (96.0%); +0.0 / 1.000
jcrew_blocker	27	27/27 (100%) [87.5–100]	16/27 (59.3%); +40.7 / <0.001	26/27 (96.3%); +3.7 / 1.000	25/27 (92.6%); +7.4 / 0.500
cuad_covenant_not_to_sue ¹	154	150/154 (97.4%) [93.5–99.0]	148/154 (96.1%); +1.3 / 0.727	149/154 (96.8%); +0.6 / 1.000	83/154 (53.9%); +43.5 / <0.001
contract_nli_explicit_id ¹	55	48/55 (87.3%) [76.0–93.7]	47/55 (85.5%); +1.8 / 1.000	46/55 (83.6%); +3.6 / 0.688	20/55 (36.4%); +50.9 / <0.001
contract_nli_notice_on_disclosure ²	70	70/71 (98.6%) [92.4–99.8]	71/71 (100%); −1.4 / 1.000	71/71 (100%); −1.4 / 1.000	36/71 (50.7%); +47.9 / <0.001
learned_hands_health ²	113	107/113 (94.7%) [88.9–97.5]	104/113 (92.0%); +2.7 / 0.375	99/113 (87.6%); +7.1 / 0.039	56/113 (49.6%); +45.1 / <0.001
cuad_liquidated_damages ²	110	105/110 (95.5%) [89.8–98.0]	107/110 (97.3%); −1.8 / 0.500	105/110 (95.5%); +0.0 / 1.000	79/110 (71.8%); +23.6 / <0.001
opp115_int_and_specific_audience ²	347	324/347 (93.4%) [90.3–95.5]	319/347 (91.9%); +1.4 / 0.302	321/347 (92.5%); +0.9 / 0.581	232/347 (66.9%); +26.5 / <0.001
All 9 tasks (combined)	949	900/949 (94.8%)	<i>combined: see Table 6.10.B</i>		

¹ Random sample, seed=42 (pre-registered in tools/random_task_pick.py). Personal jurisdiction is 28 USC §1332(a). Notice-on-compelled-disclosure full task name: contract_nli_notice_on_compelled_disclosure. Explicit-id full name: contract_nli_explicit_identification. Opp115 full name: opp115_international_and_

² Random sample, seed=43.

Cell legend. Engine column: correct/n (accuracy, 95% Wilson CI). LLM columns: correct/n (accuracy); Δ vs Engine in percentage points / exact two-sided McNemar’s p on per-case discordance with the Engine. Negative Δ indicates the LLM out-performed the Engine at that N; no negative Δ reaches significance.

When per-task results are aggregated by paired-binomial test on discordant cases across the full 949-case held-out evaluation, the Eligibility Module is significantly more accurate than each of the three frontier models at the conventional $p < 0.05$ level:

Table 6.10.B — Combined paired-binomial across 9 LegalBench tasks

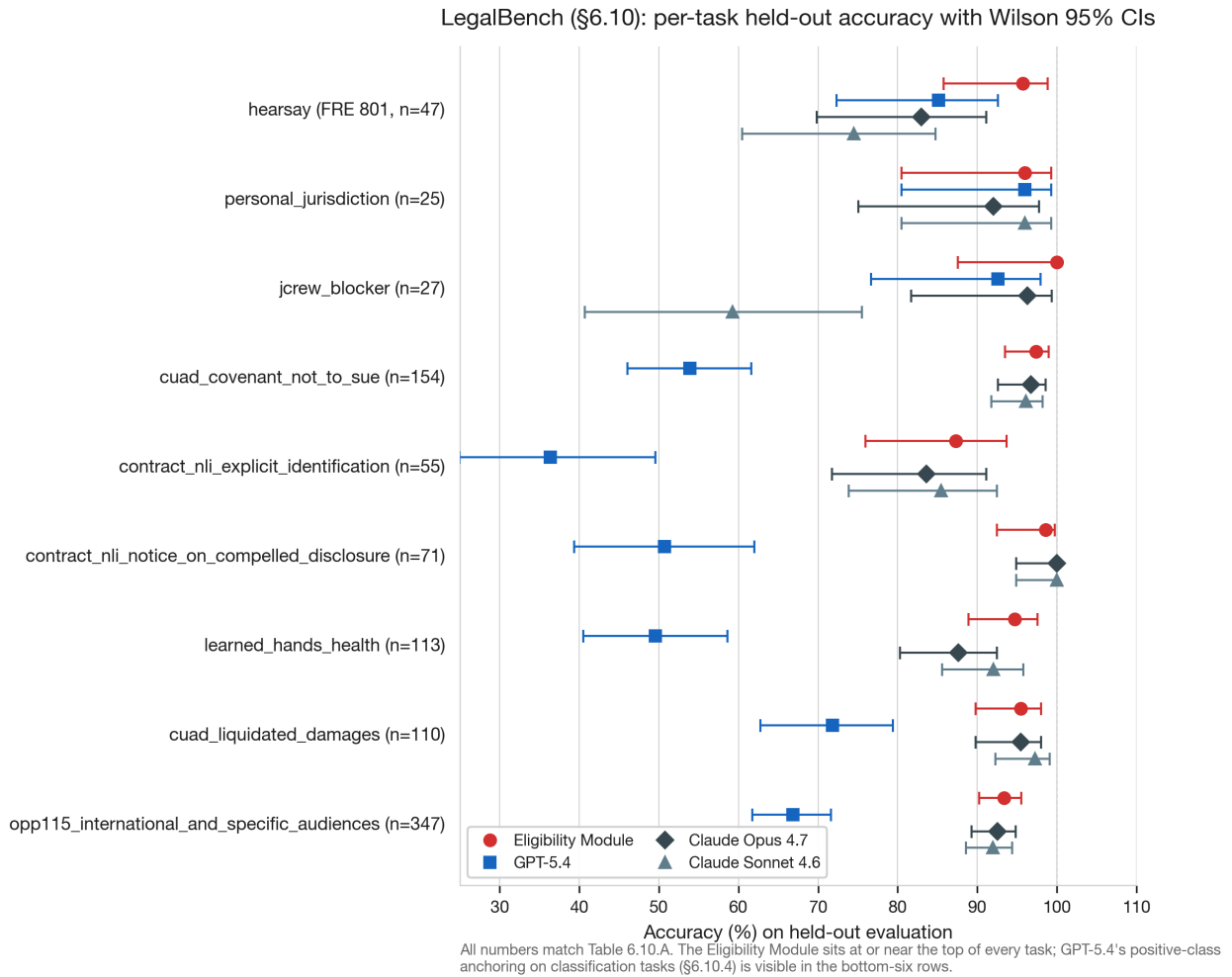
Comparison	b (Eligibility Module only correct)	c (LLM only correct)	Two-sided exact p
Module vs Sonnet 4.6	44	15	< 0.001
Module vs Opus 4.7	34	13	0.003
Module vs GPT-5.4	340	35	< 0.001

The aggregate GPT-5.4 comparison should be interpreted cautiously. Most of the discordant GPT-5.4 cases arise from the classification-prompt sensitivity analysed in §6.10.4, not from the multi-prong rule-evaluation tasks that provide the closest external analogue to the exception-chain failure pattern in §4.2. We therefore treat the GPT-5.4 aggregate as evidence that the Module remains robust under the evaluated LegalBench protocol, not as a broad claim about GPT-5.4’s best-achievable legal-reasoning performance.

Per-task statistical power. The small-N curated tasks have limited per-task power: `personal_jurisdiction` (N=25) cannot detect held-out accuracy differences smaller than approximately $\pm 20\text{pp}$ at $\alpha = 0.05$, 80% power, and `hearsay` (N=47) is similarly underpowered for $\Delta < 14\text{pp}$. The load-bearing test in this section is the combined paired-binomial across all 949 held-out cases (Table 6.10.B); per-task McNemar’s p -values are reported for completeness but should be interpreted as supplementary, particularly on the curated multi-prong tasks.

Multiple comparisons. Table 6.10.A reports 27 simultaneous per-task McNemar’s tests ($9 \text{ tasks} \times 3 \text{ LLMs}$), uncorrected. Under Bonferroni correction at $\alpha = 0.05/27 \approx 0.0019$, only the strongest comparisons remain individually significant (most of the GPT-5.4 cells; `jcrew_blocker` vs Sonnet). The combined paired-binomial in Table 6.10.B aggregates discordant pairs across all tasks and is a single test, not subject to the same correction; it is the load-bearing inferential claim and remains $p \leq 0.003$ against every model under any reasonable family-wise correction.

Figure 7: LegalBench per-task held-out accuracy with Wilson 95% CIs.



6.10.3 Selection bias and the random-sample replication

A reviewer might fairly ask whether the three “curated” tasks above (`hearsay`, `personal_jurisdiction`, `jcrew_blocker`) were chosen because they fit the Eligibility Module’s structural shape — multi-prong rule application of the kind §4.2 identifies as a frontier-LLM failure pattern. They were. To test whether the result is selection-bias-dependent, we repeated the protocol on **six additional LegalBench tasks selected by seeded random sample** from the 153-task candidate pool (the 162 upstream tasks minus the nine already integrated at pick time). The selection (seeds 42 and 43, $n=2$ and $n=4$ respectively, generated by `tools/random_task_pick.py`) was committed before the upstream Task

descriptions were inspected. A third seed (seed=44, n=2) was pre-registered at the public release tag `pre-v3.8-legalbench-preregistration` (annotated tag object `f7c5994`; the tagged commit is `58d3b5d`) but is not yet run; the tag preserves the seed-before-results discipline if those tasks are added in a future revision.

The six random-sample tasks span CUAD contract-clause classification, ContractNLI entailment, OPP115 privacy-policy classification, and LearnedHands legal-aid post categorisation — predominantly single-clause classification rather than the multi-prong rule application that §4.2 characterises as the failure pattern. Predicted-mismatch territory.

The result is consistent: the Eligibility Module wins on the random-sample tasks, but with smaller per-task margins against Sonnet 4.6 and Opus 4.7 (1–3 percentage points on the random sample, versus up to +41 percentage points on the curated multi-prong tasks); GPT-5.4’s per-task margins are larger in both arms but are dominated by the §6.10.4 prompt-format sensitivity rather than the exception-chain pattern. The combined paired-binomial test reported in Table 6.10.B includes both curated and random-sample tasks; with the curated subset removed, the random-sample combined comparison reaches significance against GPT-5.4 ($b = 331$, $c = 33$, $p < 0.001$); against Opus 4.7 it is suggestive but not significant ($b = 26$, $c = 13$, $p = 0.053$), and against Sonnet 4.6 it is positive but not significant ($b = 22$, $c = 14$, $p = 0.24$). (Correction, v3.11: earlier revisions printed random-only aggregates computed under a mislabelled five-task partition that counted `cuad_covenant_not_to_sue` as curated; the figures above are recomputed from the committed per-task discordance tables over the documented six-task random sample, and reconcile with Table 6.10.B, whose all-nine-task result is unaffected.)

The interpretation: the Eligibility Module’s structural advantage is **largest where the underlying rule has multiple genuinely-distinct prongs** combined by deterministic logic — exactly the failure pattern of §4.2. On single-clause classification tasks the structural advantage shrinks to a 1–3 percentage point edge that is directionally consistent across tasks but small per task; at the random-only level the combined test detects it clearly only against GPT-5.4 — the all-nine-task combination in Table 6.10.B is the load-bearing aggregate.

6.10.4 GPT-5.4 prompt-format sensitivity on classification tasks

GPT-5.4’s accuracy on five of the six non-curated random-sample tasks (`cuad_covenant_not_to_sue` 53.9%, `contract_nli_explicit_id` 36.4%, `contract_nli_notice_on_compelled_disclosure` 50.7%, `learned_hands_health` 49.6%, `opp115_international` 66.9%) is sharply lower than Sonnet 4.6 and Opus 4.7 on the same tasks (each clears 87% on all five). Inspection of the per-case JSON responses shows GPT-5.4 has a systematic positive-class bias on these classification tasks: on `cuad_covenant_not_to_sue` it returns “yes” on 304/308 test cases regardless of clause content (recall $\approx 100\%$, precision $\approx 50\%$). The pattern replicates on the other four tasks and is not present on the multi-prong rule tasks (`hearsay` 85.1%, `personal_jurisdiction` 96.0%, `jcrew_blocker` 92.6%).

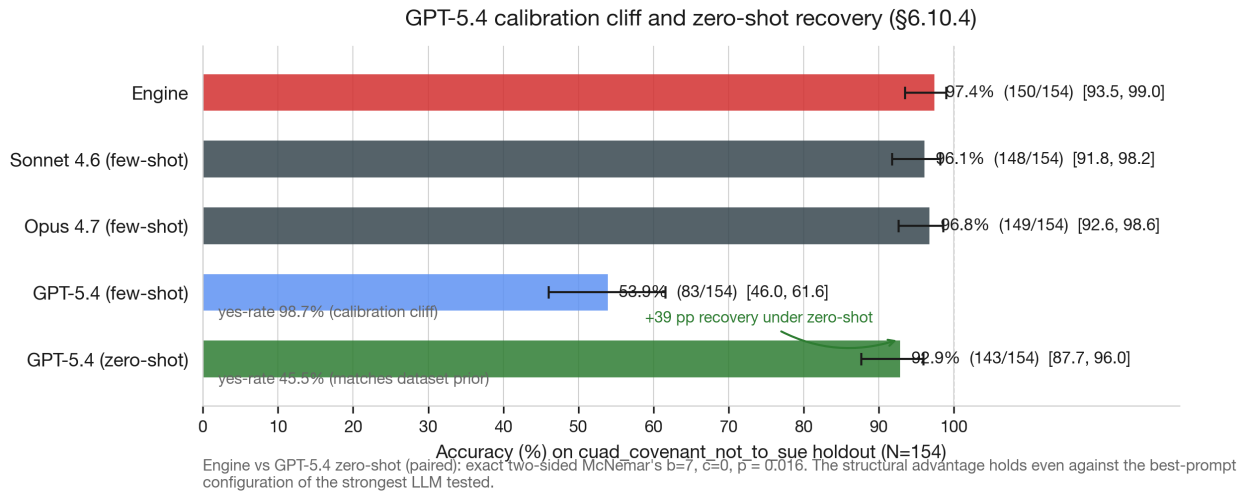
We do not characterise this as a capability statement about GPT-5.4 broadly, and we do not use these cells as primary evidence for the exception-chain claim. The pattern is specific to the LegalBench classification-style few-shot prompt format used here (the upstream canonical `base_prompt.txt`); alternative prompt strategies (chain-of-thought, structured-output, zero-shot rule-only) may recover the model’s performance.

This distinction matters because the same model recovers under a simpler prompt. We tested `cuad_covenant_not_to_sue` with a zero-shot variant: the same model and the same task description from `sources/rule.md`, but no few-shot Q/A examples — only the rule, the clause, and the question “Answer Yes or No.” On the same 154-case held-out subset, GPT-5.4 zero-shot achieves **143/154 (92.9%)**, up from 83/154 (53.9%) under the few-shot prompt — a +39pp recovery. The yes-rate normalises from 98.7% under the few-shot prompt to 45.5%, matching the dataset’s underlying $\sim 50\%$ positive prior. This empirically confirms the framing for this task: the low GPT-5.4 few-shot result is **prompt-format-coupled, not a model-capability ceiling**.

Notably, the Eligibility Module still beats GPT-5.4 zero-shot on this task: 150/154 (97.4%) vs 143/154 (92.9%), exact two-sided McNemar’s $b = 7$, $c = 0$, $p = 0.016$. The structural advantage holds even against

the best-prompt configuration of the strongest LLM tested.

Figure 8: GPT-5.4 calibration cliff and zero-shot recovery.



The relevant finding for the present paper is narrower. The multi-prong tasks support the architectural claim about compositional rule evaluation. The classification-prompt results document a separate prompt-format sensitivity in the evaluated GPT-5.4 baseline. Both are relevant to legal classification workflows, but only the former is evidence for the exception-chain failure pattern.

6.10.5 Cross-extractor: structural advantage is not extractor-specific

A second possible critique: the Eligibility Module pipeline uses an LLM extractor (Sonnet 4.6 in the headline configuration) to produce the typed field values that the engine evaluates. If the LLM-only baseline also uses Sonnet 4.6, the comparison reduces to “Sonnet split across two steps versus Sonnet doing it all in one step”, which might be a prompt-engineering artefact rather than a structural property of the Module.

To test this, we re-ran the Eligibility Module pipeline with **Opus 4.7 as the runtime extractor** on the two clearest multi-prong tasks (`hearsay` and `personal_jurisdiction`), and compared against Opus 4.7 end-to-end:

Table 6.10.C — Cross-extractor (full test split, N as shown)

Task	Sonnet 4.6 end-to-end	Sonnet-extractor + engine	Opus 4.7 end-to-end	Opus-extractor + engine
hearsay (94)	76.6%	90.4%	83.0%	86.2%
personal_jurisdiction (50)	92.0%	98.0%	92.0%	98.0%

The Eligibility Module’s structural advantage holds for **both** extractor models. On `personal_jurisdiction`, the lift is identical (+6.0pp) regardless of extractor; on `hearsay`, the lift is +13.8pp with Sonnet and +3.2pp with Opus — Opus is genuinely better at `hearsay` reasoning end-to-end, so the symbolic engine adds less incremental value to the stronger extractor. In neither case does the Eligibility Module pipeline lose to its corresponding LLM-only baseline. The result is consistent with the §4.2 framing: the deterministic conjunction step is the source of the gain, not the choice of extractor.

6.10.6 Discussion

The LegalBench results are external validation of the central claim of this paper: on multi-prong rule-evaluation tasks, the Eligibility Module’s deterministic-execution architecture is more accurate than

end-to-end LLM evaluation, and the gap is reliably detectable. The gap shrinks to a small but consistent margin on single-clause classification tasks chosen at random — the regime where there is no compositional rule structure for the symbolic engine to exploit — and the gap holds across two LLM model families (Anthropic, OpenAI) and across two choices of runtime extractor (Sonnet 4.6, Opus 4.7).

The retroactive held-out methodology is acknowledged as a methodological compromise. For the three curated tasks, hint iteration during authoring used the full LegalBench test split before the held-out partition was introduced; the held-out results in Table 6.10.A are therefore reported on a half-test subset that the authors *did* see during iteration but did not focus on. For the six random-sample tasks, the held-out methodology was applied prospectively (hints iterated only on a dev half, holdout evaluated once with frozen hints). The combined paired-binomial result in Table 6.10.B is dominated by the random-sample tasks by case count (850 of 949 cases) and remains significant.

Engine error analysis. The Eligibility Module is correct on 900/949 held-out cases (94.8%); the 49 errors are not uniformly distributed. By task: `opp115_international_and_specific_audiences` 23/347 (6.6%, the largest absolute contributor), `contract_nli_explicit_identification` 7/55 (12.7%, the highest rate), `learned_hands_health` 6/113 (5.3%), `cuad_liquidated_damages` 5/110 (4.5%), `cuad_covenant_not_to_sue` 4/154 (2.6%), `hearsay` 2/47 (4.3%), `personal_jurisdiction` 1/25 (4.0%), `contract_nli_notice_on_compelled_disclosure` 1/71 (1.4%), `jcrew_blocker` 0/27 (0.0%). The three curated multi-prong tasks contribute 3 errors out of 99 cases (3.0%); the six single-clause classification (random-sample) tasks contribute 46 errors out of 850 cases (5.4%). The pattern is consistent with the structural-advantage interpretation in §6.10.3: where the underlying rule has multiple genuinely-distinct prongs and the engine’s deterministic conjunction step is structurally aligned with the rule shape, errors are rare. Where the task is single-clause classification and the engine reduces to “the LLM extractor’s binary judgement, run through a one-clause rule,” the engine’s accuracy reflects the extractor’s accuracy on borderline cases.

Model-family and jurisdictional limitations. This evaluation tests two model families (Anthropic via Sonnet 4.6 and Opus 4.7; OpenAI via GPT-5.4). It does not test other frontier or production LLMs (Llama, Gemini, DeepSeek, Mistral). The §6.10.4 calibration finding for GPT-5.4 is shown to be prompt-format-coupled within OpenAI’s GPT-5 family but its presence or absence in other model families is untested. All nine LegalBench tasks evaluated here are US-jurisdiction English-language tasks; the §6 controlled benchmark is UK-leaning. Generalisation to other jurisdictions or non-English legal corpora is out of scope for the present revision.

What would refute this section’s central claim. A documented LLM-only configuration that, on the same 949 held-out LegalBench cases (with the deterministic seed-7 partition reproduced from `tools/test_split.py`), achieves a combined accuracy $\geq 95\%$ on the same nine tasks under any disclosed prompting strategy would substantially weaken the structural-advantage interpretation in §6.10.5–§6.10.6. Such a result would not falsify the existence of the §4.2 failure pattern (which is a small-N existence claim by design) but would show that prompt engineering alone can match the Module on this evaluation set, removing the *architectural* basis for the headline gap reported in Table 6.10.B.

The benchmark scenarios, source files, runners, statistical analysis script, and pre-registration artefacts are released in the `legalbench/` subdirectory of <https://github.com/Aethis-ai/confidently-wrong-benchmark> for replication. The pre-registration of the seed-44 random-sample replication is verifiable via the public Git tag `pre-v3.8-legalbench-preregistration` (annotated tag object `f7c5994`, tagged commit `58d3b5d`). The held-out partition reproducibility is verified at `legalbench/docs/reproducibility-checks.md` (9/9 tasks: re-running `tools/test_split.py --seed 7 --dev_fraction 0.5` from a clean checkout reproduces the index sets used in committed engine results exactly).

7. Challenges in LLM-Guided Rule Synthesis

Rule authoring is the bottleneck. The execution layer (Level 3) provides deterministic guarantees by construction, but those guarantees are only as good as the rules they operate on. Execution correctness is a necessary precondition for trust — it eliminates one entire class of error — but the hard problem remains: can an LLM reliably formalise legislation into rules that faithfully capture the legislator’s intent? This paper does not claim to solve that problem. It reports that the execution layer is robust and that we approach the Level 2 problem through two complementary quality mechanisms, summarised below. Implementation specifics — thresholds, iteration bounds, internal evaluator rubrics, and feedback-steering heuristics — are not disclosed here.

7.1 Iterative LLM-Assisted Authoring

Rules are authored through iterative synthesis with automated validation and convergence criteria: a frontier LLM generates candidate rules from authoritative source text, receives structured critique, and refines on successive passes until quality has stabilised. This is a research-engineering component, not a correctness guarantee: the purpose of Section 5.5 and Section 7.2 is to make clear which guarantees we do claim and which we do not.

7.2 Honest Assessment

These quality scores are produced by an LLM evaluator, not by independent human legal experts. They measure internal consistency against the evaluator’s rubric, not ground-truth correctness. The test-driven validation layer (Section 7.3) provides the stronger signal, but test suites are only as comprehensive as the scenarios SMEs define. This section documents engineering progress on a difficult problem; it is not a claim of solved correctness at Level 2.

7.3 Test-Driven Validation

Rule quality is gated against concrete expected outcomes, not against LLM self-evaluation alone. Subject matter experts author test suites that function as acceptance criteria for the authored rule bundle.

Test case structure. Each case pairs structured applicant data with an expected outcome (eligible / ineligible / undetermined) for a specific legal section. Cases are tagged by category — golden path, edge case, and corner case — with corner cases specifically exercising exception chains and override logic.

Validation and regression. When rules are authored or revised, the bundle is compiled and every test case is evaluated through the constraint engine. Failures produce structured diagnostics — which constraint, which expected outcome, which branch of the disjunction — that feed back into the authoring loop. A golden-case suite is preserved across rule versions so that re-authoring (due to legislation changes or quality improvements) cannot silently regress a previously-correct scenario. This outer loop, grounded in SME-defined outcomes rather than LLM self-assessment, is the primary quality gate for deployment.

8. Generalisation and Applicability

8.1 Domain-Agnostic Architecture

The system is architecturally domain-agnostic. An internal audit found that over 90% of the codebase is shared across domains. The only components requiring replacement for a new vertical are the source connectors (which fetch and parse authoritative text from domain-specific APIs) and any domain-specific business logic. The core engine, rule representation, compilation pipeline, quality evaluation, and API layer work unchanged.

8.2 Domain Injection Points

Injection Point	What It Provides	Manual or Discovered?
Source connectors	Authoritative text for the domain	Manual (one-time)
Guidance hints	Subject matter expert steering	Manual (optional)
Field registry	Typed data fields that rules reference	LLM-assisted discovery
Rules and criteria	Logical conditions compiled to constraints	LLM-generated, human-reviewable
Quality rubric	Evaluation criteria for generated rules	Domain-agnostic (built-in)

The field registry and rules are outputs of the pipeline, not inputs. The source connectors are the only early-stage domain-specific code, and they are modular: adapting to a new domain means replacing one source connector with an equivalent for another regulatory body.

8.3 Adjacent Verticals

The architecture is applicable to any domain exhibiting the three properties identified in Section 4.1: material consequence, auditability, and OR-branching logic:

- **Benefits and welfare eligibility:** Universal Credit, disability benefits, pension entitlements with complex exemption and transition rules
- **Tax compliance:** R&D tax credits, corporate tax relief pathways with safe harbours
- **Financial regulatory compliance:** MiFID II suitability, AML threshold assessments
- **Insurance claims adjudication:** Coverage determination under exclusion clauses with nested exception chains (benchmarked: Section 6.4)
- **Safety certification:** Aviation crew certification, equipment compliance with conditional overrides
- **Healthcare pathway determination:** Clinical pathway eligibility, treatment authorisation chains

9. Compliance and Regulatory Considerations

The EU AI Act (2024) classifies AI systems used in migration, asylum, and border control management as high-risk (Annex III, Category 7), requiring conformity assessments, risk management systems, and human oversight. Systems used for eligibility determination in regulated domains face increasing scrutiny under this framework. The architecture described here - deterministic execution with full provenance - is designed with these requirements in mind.

9.1 Auditability

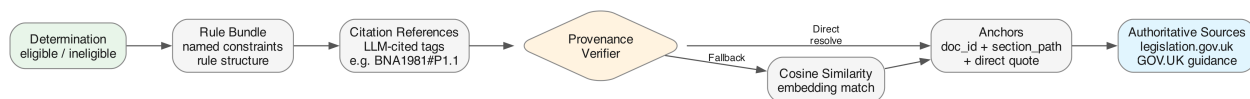
Every determination produced by the system is fully auditable through a provenance chain:

1. **Determination:** the Eligibility Module returns eligible/ineligible with the specific constraints satisfied or violated. The determination result itself carries the provenance chain: each criterion that contributed to the outcome includes its source citations (document ID, section path, and direct quote from the authoritative text). This means the provenance is not a separate lookup — it is returned alongside the determination.
2. **Rule:** each constraint maps to a named rule in the rule bundle, with its rule structure visible.
3. **Source citation:** each rule carries citations linking it to specific legislative clauses with direct quotes and section path references. During rule synthesis, the LLM’s context includes tagged source items

(e.g., [BNA1981#Schedule1/P1.1]), and the LLM cites these tags directly on each generated criterion. A provenance verifier resolves these citations against the assembled source material, creating structured anchors with document IDs, section paths, and direct quotes. When the LLM omits citations or references invalid sources, the verifier falls back to embedding-based semantic similarity as a secondary mechanism. This multi-stage citation verification with semantic fallback is more reliable than post-hoc matching alone, though the fallback path remains probabilistic and may occasionally misattribute a rule to a closely related but incorrect clause.

4. **Legislation:** the original legislative text is available via legislation.gov.uk and GOV.UK APIs.

Figure 6: Provenance Chain (Two-Stage Verification)



This chain is qualitatively different from an LLM’s reasoning trace. An LLM produces a natural language explanation that may not reflect its actual inference process. The Eligibility Module produces the exact logical path: the specific constraints evaluated, the specific values tested, and the specific branch of the OR expression that was or was not satisfied.

9.2 Explainability and Human Oversight

The Eligibility Module produces a binary determination with the exact logical path, not a confidence score. A solicitor can verify the determination by checking the logical path against their understanding of the legislation. The system supports human-supervised workflows and can be configured so that determinations are reviewed by a professional before external action is taken.

9.3 Quality Gates

The rule authoring pipeline includes multiple quality gates before a rule bundle is persisted for deployment: iterative LLM synthesis-and-critique (Section 7.1), source citation validation, and — as the strongest gate — test-driven validation against SME-defined test suites (Section 7.3). Rules must produce correct outcomes on golden path, edge case, and corner case scenarios defined by domain experts before they are deployed.

9.4 Model Drift and Bundle Certification

Frontier LLMs change between API versions and, as the v3.8 replication documents (§6.5 Finding 4), within the same nominal model alias without announcement. For systems that perform inference on every query against a frontier LLM, this raises a regulatory question with no clean answer: how can a deployed determination system be certified at time T when the underlying inference engine can change at time T+1 without notice or visibility? The Eligibility Module separates this question into two parts that can be addressed independently.

Execution layer (L3) is invariant by construction. Once a rule bundle is compiled, every determination is produced by deterministic SMT-based evaluation against that bundle. No LLM is in the inference path. Frontier model changes — silent or announced — do not alter the determinations produced by an existing bundle on existing inputs. A regulator certifying a deployed bundle is certifying a fixed artefact, not a moving target.

Authoring layers (L1 retrieval, L2 formalisation) are versioned and replayable. Each rule bundle is generated against a recorded model snapshot, with full provenance back to source legislation. If a future model would generate different rules from the same sources, the deployed bundle is unaffected; re-authoring is a deliberate, audited action that re-triggers test-driven validation (Section 7.3) and produces a new bundle version. SME test suites act as a regression gate: silent behavioural changes in the authoring model that affect rule semantics surface as test failures rather than as silent eligibility drift in production.

This separation does not eliminate the underlying drift problem in frontier models — it relocates it to the authoring boundary, where it is observable and auditable, rather than the inference boundary, where it is silent and continuous. Whether the L1/L2 boundary is itself sufficiently controlled for a given regulated workflow is a question the bundle’s quality engineering must answer (Sections 7 and 9.3); a separate open question is whether silent drift in the authoring model can produce silent incorrect rules that pass the SME test suite by coincidence — this is tracked as an open follow-up item rather than resolved within this revision.

10. Limitations and Future Work

10.1 Current Limitations

1. **Structured input only.** The benchmark tests legal reasoning accuracy, not extraction from natural language or documents. A production system requires both; the extraction layer introduces its own error surface.
2. **Level 2 validation is test-suite-dependent.** Rule formalisation quality is validated against SME-defined test suites (Section 7.3) and measured by internal LLM evaluator scores. The test suites provide ground-truth validation but are only as comprehensive as the scenarios they cover. Independent human review of generated rules against authoritative legal interpretations has not been performed.
3. **225 scenarios across four domains.** The benchmark does not exhaust legal reasoning. It tests one class of task across four domains; findings should not be generalised beyond nested exception-chain evaluation.
4. **Model versioning.** Results are tied to specific model versions available at time of evaluation. The public benchmark dataset allows this to be re-evaluated as models improve.
5. **Benchmark performance is not a production guarantee.** The accuracy figures reported here reflect performance on fixed benchmark scenarios under controlled conditions. They do not establish correctness for any specific real-world legal determination. Production deployment of an automated eligibility or compliance system requires professional legal review per jurisdiction, ongoing monitoring of model and legislative changes, and appropriate human oversight. Nothing in this paper should be read as legal advice or as a warranty of fitness for a particular legal purpose.
6. **The failure mode is bounded.** The systematic failures characterised here concentrate at three-level exception depth and on the multi-prong compositional sub-task (the precise scope phrase introduced in §1: *multi-prong compositional rule evaluation under nested exceptions*). On shallower or non-exception eligibility logic, frontier-model accuracy on this benchmark is high, and §6.4.1 documents a declining adversarial failure rate as frontier models update. The contribution is the demonstration of a durable boundary condition where deterministic execution is required, not a claim of broad LLM unreliability.

10.2 Future Work

Priority	Item
High	Human evaluation calibration: compare LLM quality scores against human-labelled rule sets
High	Multi-section regression suite: automated quality runs across all sections
Medium	Natural language input testing: extend benchmark to unstructured applicant narratives

Priority	Item
Medium	Multi-model evaluation ensemble: cross-validate quality scores across model families
Low	Independent legal review: solicitor verification of expected values

Acknowledgments

We thank an anonymous reviewer (referee report received 2026-04-28) for the framing of the architectural contribution as a *relocation* of uncertainty rather than its elimination. That framing clarified the rhetorical centre of gravity of §2, §5.5, and §9.4, and produced the boundary-condition restatement of the failure mode in §10.1. We also thank the project’s adversarial-review process — both human and automated — for sustained pressure on the specification-vs-execution distinction, the prompt-engineering claim, and the scope of the failure regime characterised here.

References

- Bench-Capon, Trevor, and Henry Prakken. 2010. “Using Argument Schemes for Hypothetical Reasoning in Law.” *Artificial Intelligence and Law* 18 (2): 153–74.
- Dziri, Nouha, Ximing Lu, Melanie Sclar, Xiang Lorraine Li, et al. 2023. “Faith and Fate: Limits of Transformers on Compositionality.” *Advances in Neural Information Processing Systems (NeurIPS)*.
- European Parliament. 2024. *Regulation (EU) 2024/1689 Laying down Harmonised Rules on Artificial Intelligence (AI Act)*. Official Journal of the European Union.
- Gao, Luyu, Aman Madaan, Shuyan Zhou, et al. 2023. “PAL: Program-Aided Language Models.” *Proceedings of ICML*.
- Garcez, Artur d’Avila, Luís C. Lamb, and Dov M. Gabbay. 2009. *Neural-Symbolic Cognitive Reasoning*. Springer.
- Governatori, Guido, and Antonino Rotolo. 2010. “Changing Legal Systems: Legal Abrogations and Annulments in Defeasible Logic.” *Logic Journal of the IGPL* 18 (1): 157–94.
- Guha, Neel, Julian Nyarko, Daniel E. Ho, et al. 2023. “LegalBench: A Collaboratively Built Benchmark for Measuring Legal Reasoning in Large Language Models.” *arXiv Preprint arXiv:2308.11462*.
- Han, Simeng, Hailey Schoelkopf, Yilun Zhao, Zhenting Qi, Martin Riddell, et al. 2022. “FOLIO: Natural Language Reasoning with First-Order Logic.” *arXiv Preprint arXiv:2209.00840*.
- Kambhampati, Subbarao, Karthik Valmeekam, Lin Guan, et al. 2024. “Position: LLMs Can’t Plan, but Can Help Planning in LLM-Modulo Frameworks.” *Proceedings of the 41st International Conference on Machine Learning (ICML)*.
- Kojima, Takeshi, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. “Large Language Models Are Zero-Shot Reasoners.” *Advances in Neural Information Processing Systems*

(*NeurIPS*).

- Marcus, Gary. 2020. “The Next Decade in AI: Four Steps Towards Robust Artificial Intelligence.” *arXiv Preprint arXiv:2002.06177*.
- McCarty, L. Thorne. 1977. “Reflections on TAXMAN: An Experiment in Artificial Intelligence and Legal Reasoning.” *Harvard Law Review* 90: 837–93.
- Mirzadeh, Iman, Keivan Alizadeh, Hooman Shahrokhi, Oncel Tuzel, Samy Bengio, and Mehrdad Farajtabar. 2024. “GSM-Symbolic: Understanding the Limitations of Mathematical Reasoning in Large Language Models.” *arXiv Preprint arXiv:2410.05229*.
- Mohun, James, and Alex Roberts. 2020. *Cracking the Code: Rulemaking for Humans and Machines*. OECD Working Papers on Public Governance No. 42. OECD. <https://doi.org/10.1787/3afe6ba5-en>.
- Mowbray, Andrew, Philip Chung, and Graham Greenleaf. 2023. “Representing Legislative Rules as Code: Reducing the Problems of ‘Scaling up’.” *Computer Law & Security Review* 48: 105799. <https://doi.org/10.1016/j.clsr.2022.105799>.
- Nute, Donald. 1994. “Defeasible Logic.” In *Handbook of Logic in Artificial Intelligence and Logic Programming*, vol. 3. Oxford University Press.
- Olausson, Theo X., Alex Gu, Benjamin Lippert, et al. 2023. “LINC: A Neurosymbolic Approach for Logical Reasoning by Combining Language Models with First-Order Logic Provers.” *arXiv Preprint arXiv:2310.15164*.
- Pan, Liangming, Michael Saxon, Wenda Xu, Deepak Nathani, Xinyi Wang, and William Yang Wang. 2023. “Logic-LM: Empowering Large Language Models with Symbolic Solvers for Faithful Logical Reasoning.” *arXiv Preprint arXiv:2305.12295*.
- Reiter, Raymond. 1980. “A Logic for Default Reasoning.” *Artificial Intelligence* 13 (1–2): 81–132.
- Sergot, Marek, Fariba Sadri, Robert Kowalski, Frank Kriwaczek, Peter Hammond, and H. T. Cory. 1986. “The British Nationality Act as a Logic Program.” *Communications of the ACM* 29 (5): 370–86.
- Shi, Freda, Xinyun Chen, Kanishka Misra, et al. 2023. “Large Language Models Can Be Easily Distracted by Irrelevant Context.” *Proceedings of ICML*.
- Valmeekam, Karthik, Alberto Olmo, Sarath Sreedharan, and Subbarao Kambhampati. 2022. “Large Language Models Still Can’t Plan (a Benchmark for LLMs on Planning and Reasoning about Change).” *arXiv Preprint arXiv:2206.10498*.
- Valmeekam, Karthik, Kaya Stechly, and Subbarao Kambhampati. 2024. “LLMs Still Can’t Plan; Can LRMs? A Preliminary Evaluation of OpenAI’s O1 on PlanBench.” *arXiv Preprint arXiv:2409.13373*.
- Wang, Xuezhi, Jason Wei, Dale Schuurmans, et al. 2023. “Self-Consistency Improves Chain of Thought Reasoning in Language Models.” *ICLR*.
- Wei, Jason, Xuezhi Wang, Dale Schuurmans, et al. 2022. “Chain-of-Thought Prompting Elicits Reasoning in Large Language Models.” *Advances in Neural Information Processing Systems (NeurIPS)*.

Technical Appendix A: Benchmark Methodology

A.1 Life in the UK Scenario Matrix

Field	Type	Values Tested
test_passed	Boolean	True, False
medical_exempt	Boolean	True, False
discretion_applied	Boolean	True, False
applicant_age	Integer	15, 17, 18, 30, 64, 65, 70

Total combinations: $2 \times 2 \times 2 \times 7 = 56$ scenarios

Expected value formula: $\text{eligible} = \text{test_passed} \text{ OR } (\text{age} < 18) \text{ OR } (\text{age} \geq 65) \text{ OR } \text{medical_exempt} \text{ OR } \text{discretion_applied}$

A.2 English Language Scenario Design

43 scenarios (24 baseline + 19 adversarial): MESC nationality route; Canada bilingual special case; SELT expiry rules; degree certification; near-miss combinations; multi-field distractors; age boundary stacking.

A.3 Spacecraft Scenario Design

68 scenarios (48 baseline + 20 adversarial) using a synthetic statute modelled on UK legislative structure. Exercises eight patterns: UNSAT early termination, multi-field AND, multi-route OR, three-level exception chain, nested IMPLIES, ENUM + IN, compound multi-requirement, adversarial traps (veteran age-independence, IMPLIES vacuous truth, direct satisfaction vs. exemption).

A.4 Construction Insurance Scenario Design

58 scenarios using a synthetic CAR policy defect exclusion endorsement modelled on London market DE3/DE5 clause structure. Exercises eight patterns: UNSAT early termination, multi-field AND, multi-route OR, five-level exception chain, nested IMPLIES, ENUM + IN, DATE-bounded BOOL, absolute exclusion, plus demo, adversarial, and boundary-stacking scenarios.

Fields:

Field	Type	Values
car.policy.period_valid	Boolean	True, False
car.property.category	Enum	permanent_works, temporary_works, plant_equipment, existing_structures, materials_on_site
car.loss.is_physical	Boolean	True, False
car.component.is_defective	Boolean	True, False
car.defect.origin	Enum	design, specification, materials, workmanship, none
car.claim.is_rectification	Boolean	True, False
car.claim.is_access_damage	Boolean	True, False
car.damage.consequence_of_fault	Boolean	True, False
car.project.value_millions_gbp	Integer	1–800
car.notification.within_period	Boolean	True, False
car.contract.jct_compliant	Boolean	True, False
car.defect.was_known_prior	Boolean	True, False
car.defect.has_engineer_assessed	Boolean	True, False

Field	Type	Values
<code>car.component.installed_by_s</code>	Boolean	True, False

A.5 LLM Prompt Structure

1. Full section text (markdown-formatted legislation and guidance)
2. Structured applicant data as key-value pairs
3. System prompt: *"You are a legal eligibility assessor. Given the legislation and applicant data, determine whether the applicant meets the requirement. Respond with JSON: {"eligible": true/false, "reasoning": "..."}"*

No special instructions about exemptions, alternative routes, or OR-branching.

A.6 Reproduction

```
# Full benchmark (225 scenarios across all sections)
uv run python scripts/benchmark_accuracy_ab.py --model claude-opus-4-6

# Single section
uv run python scripts/benchmark_accuracy_ab.py --model gpt-5.4 --section spacecraft

# Construction insurance comparison
uv run python scripts/llm_vs_engine_comparison.py --models "gpt-5.4,gpt-4.1-mini"

# More LLM runs for higher confidence
uv run python scripts/benchmark_accuracy_ab.py --model gpt-5-mini --llm-runs 5

# Parallel evaluation (faster for large scenario counts)
uv run python scripts/benchmark_accuracy_ab.py --model gpt-5.4 --concurrency 10
```

Supported models: gpt-5.4, gpt-5.3, gpt-5.2, gpt-5-mini, gpt-5-nano, gpt-4.1-mini, claude-opus-4-7, claude-opus-4-6, claude-sonnet-4-6, claude-haiku-4-5, gemini-3.1-pro, gemini-3.1-flash-lite, deepseek-r1.

Technical Appendix B: Architecture Overview

The Eligibility Module uses a typed rule representation that is parsed from a constrained DSL, compiled to formal constraints, and evaluated by an SMT-based constraint evaluation engine. Rules are organised into groups representing alternative legal routes (OR'd within groups, AND'd across groups), with support for custom outcome logic including material implication.

The compilation pipeline transforms authored rules into a live constraint evaluation engine capable of deterministic evaluation, dynamic question generation, and provenance tracking from determination back to source legislation.

All rules, fields, and rule sets are stored immutably with versioning and supersedes references. This architecture detail is available under NDA for technical due diligence.

Author Contributions

P.S. conceived the paper, designed and implemented the Eligibility Module — including the code-generation layer, compilation architecture, SMT constraint-evaluation engine, rule-authoring tools, and benchmark harness — designed and ran all experiments, analysed the results, and drafted the manuscript. P.S. additionally designed and ran the §6.10 LegalBench external-validation programme: per-task rule-bundle authoring, SME-guidance design, runtime extractor configurations, statistical-analysis pipeline (`tools/significance.py`), held-out partition methodology, pre-registered random sampling protocol, GPT-5.4 zero-shot prompt-sensitivity check, and Figures 7 and 8. J.K. proposed the use of an SMT solver as the execution engine for deterministic rule evaluation within P.S.’s existing code-generation and compilation architecture, and contributed ongoing design discussion. L.D., Immigration Solicitor and accredited Senior Caseworker and Supervisor under the Law Society Immigration and Asylum Accreditation Scheme, with over twenty years’ experience in immigration law, authored the benchmark scenarios for the British Nationality Act eligibility determinations (`life_uk` and `english_language` sections) and validated their legal formalisations; she did not contribute to the spacecraft, construction insurance, or LegalBench materials. All authors reviewed the final manuscript.

Competing Interests

All authors are affiliated with Aethis, which develops and commercially deploys the Eligibility Module evaluated in this paper. The benchmark scenarios, harness, prompts, and per-case result artefacts are publicly released (see Data and Code Availability) so that all reported comparisons can be independently reproduced.

Data and Code Availability

The benchmark scenarios, expected outcomes, prompts, and evaluation harness for the §6.1–§6.9 controlled benchmark (225 paper-scope scenarios across four domains) are released publicly as the *confidently-wrong-benchmark* repository: <https://github.com/Aethis-ai/confidently-wrong-benchmark>.

The §6.10 LegalBench external-validation harness (15 per-task projects covering the 9 reported tasks plus 6 supplementary, including verbatim CC-BY-4.0 source-rule files, all per-case engine + LLM result JSONs for every reported number, the dev/holdout partition utility, the seeded random-sample task picker, the per-task statistical-analysis pipeline, and the seed-44 pre-registration tag) is released as the `legalbench/` subdirectory of the same repository: <https://github.com/Aethis-ai/confidently-wrong-benchmark/tree/main/legalbench>. The migration is tagged as `pre-v3.8-legalbench-preregistration` (annotated tag object `f7c5994`, tagged commit `58d3b5d`) — that tag is the externally-verifiable boundary for the §6.10.3 random-sample-replication claim. Held-out partition reproducibility is verified at `legalbench/docs/reproducibility-checks.md` (9/9 tasks reproduce exactly from a clean checkout).

Both sub-corpora are intended to allow independent re-evaluation of the reported results and evaluation of future models against the same task structure.

Version 3.11.0 · Working paper · July 2026

Changelog: v3.11.0 — consistency and staleness pass; no benchmark numbers changed; added a zero-context cold-read verification pass (16 findings triaged: fixed / false-positive / number-disputed); corrected the §6.10.3 random-only combined aggregates, previously computed under a mislabelled five-task partition — recomputed from the committed per-task discordance tables over the documented six-task random sample (significance vs Sonnet 4.6 at the random-only level does not survive the correction; Table 6.10.B is unaffected). Unified the R2 false-negative accounting to the committed 7→4 figure across §2, §3.4, §6.7 R2 and Figure 5; scoped Finding 2’s no-false-positives claim to the March 2026 baseline evaluations (§6.3–§6.4)

and cross-referenced the §6.10 LegalBench positive-class anchoring; corrected Finding 3 against Table 2; reconciled the §6.1 / Table 9 temperature description (added the $T=0.3$ sensitivity-control note); re-dated the §6.9 pre-registration commitment to a subsequent revision with a July 2026 status note; distinguished the pre-registration annotated-tag object from its tagged commit SHA (§6.10.3, Data and Code Availability); added dated (evaluation-time) qualifiers to the previously-undated “strongest Anthropic model” claims; extended the §6.1 model roster and evaluation-coverage matrix to all evaluated configurations (added Claude Opus 4.7 and GPT-4.1-mini) and updated the §6 heading to eight LLMs; and added a Competing Interests statement. v3.10.0 — added a §3.1 paragraph on the “Rules as Code” movement (Mohun & Roberts 2020, Mowbray et al. 2023), positioning decision-tree legislative modelling as sharing the paper’s core premise (formal encoding over statistical inference) while distinguishing the constraint-based approach by its ability to prove properties — completeness, consistency, unreachability of dead states — that decision-tree representations do not natively support. Two references added to the bibliography. No benchmark numbers changed; no downstream commercial document needs to re-sync. Earlier revision history omitted; full version history is tracked in git.